# Distributional Determinantal Point Process for Repulsive Clustering of Distributions

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**Abstract**

We introduce the distributional determinantal point process (dDPP) as a novel repulsive point process whose atoms are probability distributions rather than points in a real space. The dDPP is constructed via an L-ensemble with a sliced Wasserstein (SW) kernel between distributions. We show its validity as a well-defined point process. In the discrete setting, we derive concentration results for plug-in estimators of the L-ensemble, the correlation kernel, and their determinants given i.i.d. samples from the distributional atoms. Leveraging this framework, we propose a distribution-valued random partition model by way of a repulsive generalized Bayesian mixture model. The model places a dDPP prior over the atoms of the mixing measure and defines a generalized likelihood based on SW distance. To summarize posterior inference, we develop a decision-theoretic approach to report a point estimate of the mixing measure as a Bayes rule under a hierarchical optimal transport utility function. The latter is a natural choice given that the mixing measure is itself a distribution over distributions. We use the proposed framework for inference with single-cell gene expression data and human epilepsy data, producing interpretable and well-separated clusters that reflect meaningful structure in the data.

*Keywords:* repulsive prior, sliced optimal transport, sliced Wasserstein kernel, generalized Bayes, random partitions

# 1 Introduction

We propose the distributional determinantal point process (dDPP) as a novel repulsive point process with distribution-valued atoms. The construction leverages a sliced Wasserstein (SW) kernel (Kolouri et al. 2016, Carrière et al. 2017) between distributions to build an L-ensemble. We use the dDPP to propose a distribution-valued random partition model by way of setting up a mixture model with (i) a dDPP prior over the atoms of the mixing measure and, (ii) a generalized likelihood (Bissiri et al. 2016, Chakraborty et al. 2025) based on SW distance (Rabin et al. 2012, Bonneel et al. 2015, Nguyen 2025).

Probabilistic clustering (random partition) is often implemented by way of discrete mixture models. Latent allocation variables associate each observation with a specific mixture component. Interpreting these latent variables as cluster membership indicators defines the desired clusters. Specifying a prior over the partition in this setup reduces to placing a prior over the mixing measure that governs the mixture. Under symmetry conditions known as exchangeability, every random partition admits such a characterization based on a prior over the discrete mixing measure (Kingman 1978). Prior distributions on random probability measures are known as nonparametric Bayesian models. Prominent examples are the Ferguson-Dirichlet process (DP) (Ferguson 1973), the Pitman–Yor process (Pitman & Yor 1997), and the wider family of normalized completely random measures (Lijoi et al. 2005, 2007).

A common problem of the mentioned models is the tendency to produce poorly separated mixture components, which can undermine the interpretability of the resulting clusters. The problem is mitigated by asymptotic results for large samples (Rousseau & Mengersen 2011), or can be avoided by repulsive priors (Beraha et al. 2022, Cremaschi et al. 2025, Petralia et al. 2012, Xie & Xu 2020, Pedroso et al. 2026). Placing a repulsive point process prior on the atoms of the mixing measure, i.e., the mixture locations, effectively pushes components of the mixture model apart. In particular, the determinantal point process (DPP) (Hough et al. 2009, Macchi 1975, Kulesza & Taskar 2012, Lavancier et al. 2015) has been widely used as a prior in repulsive mixture models (Bianchini et al. 2020, Xu et al. 2016). Recently, Beraha et al. (2025) and Song et al. (2025) proposed a unified framework for analyzing the associated mixing measure and characterizing the distribution of the DPP prior both a priori and a posteriori. In this paper, we follow a similar strategy to develop clustering for distribution-valued data by way of a novel distribution-valued dDPP.

Alternatively, multilevel clustering (Ho et al. 2017) can also be viewed as a method for finding partitions of distributions. A K-means algorithm and a K-centers algorithm in the Wasserstein space are proposed in Zhuang et al. (2022) and Okano & Imaizumi (2025), respectively. For Bayesian approaches, nested DP mixture models (Rodriguez et al. 2008) provide two levels of clustering, including clustering at the distributional level. A common challenge in this literature is the high computational complexity arising from the rich structure of the atoms (distributions). In particular, the Wasserstein distance is computationally expensive, with super-cubic complexity (Peyré & Cuturi 2019). Bayesian nonparametric models such as the nested DP require a full specification of the generative model for tractable posterior inference, tend to produce poorly separated clusters, and involve computationally expensive implementation of posterior inference. Adding repulsiveness into the prior poses both theoretical and computational challenges.

To the best of our knowledge, no prior work has considered repulsive priors with distributions as atoms or derived associated mixture models. Despite the inherent challenges, repulsive mixture models for distributional data are essential in practice when interpretable inference is required. We consider two typical examples as motivating applications. In one example, we aim to cluster donors based on single-cell gene expression data, which can be treated as (empirical) distributions (over cells) of gene expression. Another example is clustering of epilepsy patients based on their history of seizures, which can be cast as a distribution over windows of observed binary repeat measurements. Inference in both examples involves clustering of distributions. We implement the desired model-based clustering by the proposed extension of DPP mixture models to distribution-valued data.

We set up dDPP using a SW kernel for distributions. The choice of the SW kernel is particularly appealing for this construction because, first, it endows the construction that guarantees the dDPP to be a well-defined point process, unique property that many alternative distance functions for distributions such as the Wasserstein distance do not possess. Second, in the discrete setting where computation is more tractable, concentration results for plug-in estimators of both the L-ensemble kernel and the correlation kernel of the DPP construction allow for the use of empirical distributions as practical proxies for distributional atoms. By placing a dDPP prior over the atoms of the mixing measure and adopting SW distance to construct a generalized likelihood (Bissiri et al. 2016), the resulting repulsive mixture model favors well-separated clusters without requiring a fully specified generative model, substantially broadening its applicability.

The remainder of the article is organized as follows. Section 2 reviews SW distance and SW kernel that underpin our construction. In section 3, we introduce the dDPP, establish its well-definedness under a compactness assumption, and derive concentration results for plug-in estimates (using empirical distributions) of the L-ensemble and correlation kernels in the discrete setting. Section 4 introduces a generalized repulsive Bayesian mixture model. Section 5 includes results on posterior consistency, posterior characterization, marginal Markov chain Monte Carlo inference, and a decision-theoretic framework to summarize the posterior random partition. In section 6, we demonstrate the proposed framework with inference for single-cell gene expression data and human epilepsy data, showing that the repulsive prior yields interpretable and well-separated clusters. We conclude with a final discussion in section 7. Technical proofs and additional experimental results are provided in the Supplementary Materials.

A brief note on notation. The Dirac delta measure concentrated at a point $x$ is written as $\delta_x$. For any integer $d \geq 2$, the set $\mathbb{S}^{d-1} = \{\theta \in \mathbb{R}^d : \|\theta\|_2 = 1\}$ denotes the unit hypersphere in $\mathbb{R}^d$. When comparing two sequences $a_n$ and $b_n$, notation $a_n = \mathcal{O}(b_n)$ means that there is an absolute constant $c > 0$ satisfying $a_n \leq c b_n$ for every $n \geq 1$. Let $(\mathcal{X}_1, \Sigma_1)$ and $(\mathcal{X}_2, \Sigma_2)$ be measurable spaces, and suppose $f : \mathcal{X}_1 \to \mathcal{X}_2$ is a measurable map. For a measure $\mu$ on $(\mathcal{X}_1, \Sigma_1)$, the push-forward measure $f\sharp\mu$ on $(\mathcal{X}_2, \Sigma_2)$ is given by $f\sharp\mu(B) = \mu\left(f^{-1}(B)\right), \quad \forall B \in \Sigma_2$. The standard $k$-simplex is denoted $\Delta^k$. A vector $(\pi_1, \ldots, \pi_k) \in \Delta^k$ satisfies $\pi_k \geq 0$ for each $k = 1, \ldots, k$ together with $\sum_{k=1}^k \pi_k = 1$. Finally, $\mathcal{P}_2(\mathbb{R}^d)$ denotes the set of all distributions supported on $\mathbb{R}^d$ with finite second moment. Further notation is introduced as needed.

# 2 Sliced Wasserstein Distance and Kernel

By way of a brief review of Wasserstein distance, sliced Wasserstein distance, and sliced Wasserstein kernel, we introduce some notation and definitions. For $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$, Wasserstein distance (Villani 2009) between $\mu$ and $\nu$ is defined as follows:

$$\mathrm{W}_2^2(\mu, \nu) = \inf_{\pi \in \Pi(\mu,\nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|_2^2 \, \mathrm{d}\pi(x, y),$$

where $\Pi(\mu, \nu) = \left\{\pi \in \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d) \mid \pi(A \times \mathbb{R}^d) = \mu(A),\ \pi(\mathbb{R}^d \times B) = \nu(B)\ \forall A, B \subset \mathbb{R}^d\right\}$ is the set of all transportation plans/couplings. While being geometrically meaningful, Wasserstein distance is well-known to be computational expensive in practice. For continuous cases, Wasserstein distance is usually intractable, except for some special cases like Gaussians and the univariate setting. For discrete cases, the time complexity of Wasserstein distance is $\mathcal{O}(m^3 \log m)$ (Peyré et al. 2019) with $m$ being the maximum number of atoms of two distributions. Therefore, it is burdensome to use it for large discrete distributions.

One of the solutions to avoid the computational issue of Wasserstein distance is sliced Wasserstein (SW) distance (Rabin et al. 2014, Nguyen 2025). SW distance between two distributions $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$ is defined as:

$$\mathrm{SW}_2^2(\mu,\nu) = \mathbb{E}_{v\sim\mathcal{U}(\mathbb{S}^{d-1})}[W_2^2(P_v\sharp\mu, P_v\sharp\nu)],$$

where $\mathcal{U}(\mathbb{S}^{d-1})$ is the uniform distribution over the unit hypersphere in $d$ dimension, and $P_v\sharp\mu$ and $P_v\sharp\nu$ denote the pushforward distribution of $\mu$ and $\nu$ through the function $P_v(x) = v^\top x$. Utilizing closed-form expression for one-dimensional Wasserstein distance, we rewrite SW as:

$$\mathrm{SW}_2^2(\mu,\nu) = \mathbb{E}_{v\sim\mathcal{U}(\mathbb{S}^{d-1})}\left[\int_0^1 (F^{-1}_{P_v\sharp\mu}(t) - F^{-1}_{P_v\sharp\nu}(t))^2 \mathrm{d}t\right], \tag{1}$$

where $F^{-1}_{P_v\sharp\mu}(t)$ and $F^{-1}_{P_v\sharp\nu}(t)$ are quantile functions of $P_v\sharp\mu$ and $P_v\sharp\nu$, respectively. When $\mu$ and $\nu$ are discrete distributions with at most $m$ atoms, evaluating $\int_0^1 (F^{-1}_{P_v\sharp\mu}(t) - F^{-1}_{P_v\sharp\nu}(t))^2 \mathrm{d}t$ has the time complexity of only $\mathcal{O}(m\log m)$. The expectation in (1) is often intractable, and hence, numerical approximation is needed (Nguyen et al. 2024, Leluc et al. 2024), for example via Monte Carlo (Bonneel et al. 2015):

$$\widehat{\mathrm{SW}}_2^2(\mu,\nu;V) = \frac{1}{V}\sum_{l=1}^{V} \mathrm{W}_2^2(P_{v_l}\sharp\mu, P_{v_l}\sharp\nu),$$

where $v_1,\ldots,v_V \overset{i.i.d.}{\sim} \mathcal{U}(\mathbb{S}^{d-1})$ with $V$ being the number of Monte Carlo samples or the number of projections. The overall time complexity of this approximation is $\mathcal{O}(Vm\log m)$.

Beyond the computational benefit, SW has an unique property compared to Wasserstein distance. In particular, SW is *Hilbertian* while Wasserstein distance is not, implying in particular that we can use SW to build a positive definite kernel needed for DPP. In more detail, there exists a mapping $\Psi : \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{L}_2(\mathbb{S}^{d-1}\times\mathbb{R})$, such that:

$$\mathrm{SW}_2^2(\mu,\nu) = \|\Psi(\mu) - \Psi(\nu)\|^2_{\mathbb{L}_2(\mathbb{S}^{d-1}\times\mathbb{R})},$$

where $\|\cdot\|_{\mathbb{L}_2(\mathbb{S}^{d-1}\times\mathbb{R})}$ is the functional norm. From (1), one example of the mapping is $\Psi[\mu](v,t) = F^{-1}_{P_v\sharp\mu}(t)$ (there are alternative mappings – see examples in Kolouri et al. (2016)). With the Hilbertian property, SW distance results in a valid kernel between distributions, which is defined as follows:

$$\mathrm{SWK}(\mu,\nu) = \exp\Big[-\gamma\,\mathrm{SW}_2^2(\mu,\nu)\Big], \tag{2}$$

where $\gamma > 0$ is a scale (bandwidth) parameter. The above SW kernel is positive definite (Kolouri et al. 2016, Carrière et al. 2017, Meunier et al. 2022). Importantly, replacing $\mathrm{SW}_2^2(\mu,\nu)$ by its numerical approximation $\widehat{\mathrm{SW}}_2^2(\mu,\nu;V)$ still defines a positive definite kernel. Finally, we note that there are also other alternative definitions of SW-based kernels (Kolouri et al. 2016, Luong et al. 2025). However, we will focus on (2).

# 3 Distributional Determinantal Point Processes

## 3.1 A Distribution-Valued Determinantal Point Processes (dDPP)

We introduce distributional determinantal point processes (dDPP) as a DPP on the set of distributions $\Theta \subset \mathcal{P}_2(\mathbb{R}^d)$. We equip $\Theta$ with SW topology and its Borel $\sigma$-algebra $\mathcal{B}(\Theta)$. In the construction, we use a base measure $P_0$ on $(\Theta, \mathcal{B}(\Theta))$. As an example, $\Theta$ can be the subset of finitely supported probability measures (distributions):

$$\Theta = \left\{ \nu = \sum_{j=1}^{m} \alpha_j \delta_{x_j} : m \in \mathbb{N}, \alpha_j > 0, x_j \in \mathbb{R}^d, \sum_{j=1}^{m} \alpha_j = 1 \right\},$$

and $P_0$ may be taken as a normalized random measure (NRM) (Regazzini et al. 2003). As another example, $\Theta$ can be a finite set of distributions

$$\Theta = \{\nu_1 \ldots, \nu_Q\} \subset \mathcal{P}_2(\mathbb{R}^d),$$

where each $\nu_i$ is a fixed probability measure on $\mathbb{R}^d$. In this discrete setting, $P_0$ is naturally taken as a probability measure on the finite set $\Theta$, e.g., the uniform measure $P_0 = \frac{1}{Q}\sum_{i=1}^{Q}\delta_{\nu_i}$. With a base measure $P_0$, we now define dDPP through a L-ensemble kernel.

**Definition 1** (dDPP)**.** *A distributional determinantal point process (dDPP) on $\Theta$ with L-ensemble kernel $L$ between distributions and base probability measure $P_0$ is a point process $\Phi$ whose Janossy density with respect to $P_0^{\otimes k}$ is:*

$$j_k(\nu_1,\ldots,\nu_k) \propto \det(L(\nu_i,\nu_j))_{i,j=1}^k \tag{3}$$

*for every $k \geq 1$ and every unordered finite configuration $\{\nu_1,\ldots,\nu_k\} \subset \Theta$ with normalizing constant*

$$\sum_{k=0}^{\infty}\frac{1}{k!}\int_{\Theta^k}\det\Big(L(\nu_i,\nu_j)\Big)_{i,j=1}^k \mathrm{d}P_0(\nu_1)\ldots\mathrm{d}P_0(\nu_k).$$

Next, we prove that under a compactness assumption on $\Theta$, the dDPP in Definition 1 with SW kernel (2) for $L(\nu_i,\nu_j)$ is a well-defined DPP. In particular, beyond showing that $L(\cdot,\cdot)$ is symmetric positive definite, we show that the normalization constant is finite so that (3) is a well-defined density.

**Assumption 1** (Compactness of $\Theta$ under $\mathrm{SW}_2$)**.** *We assume that $\Theta \subset \mathcal{P}_2(\mathbb{R}^d)$ is compact under the sliced Wasserstein metric ($SW_2$). This holds if the following two conditions are satisfied:*

1. *Uniform tightness. For every $\varepsilon > 0$, there exists a compact set $\mathcal{K} \subset \mathbb{R}^d$ such that*

$$\nu(\mathcal{K}) \geq 1-\varepsilon \quad \textit{for all } \nu \in \Theta.$$

2. *Uniform integrability of second moments. The second moments are uniformly integrable over $\Theta$:*

$$\lim_{R\to\infty}\sup_{\nu\in\Theta}\int_{\|x\|\geq R}\|x\|^2\mathrm{d}\nu(x) = 0.$$

Under the conditions in Assumption 1, $\Theta$ is compact in $(\mathcal{P}_2(\mathbb{R}^d), \mathrm{W}_2)$ (Villani 2009). Since $\mathrm{SW}_2$ metrizes weak convergence on $\mathcal{P}_2(\mathbb{R}^d)$ (Nadjahi et al. 2020), and weak convergence together with convergence of second moments is equivalent to $W_2$-convergence, compactness under $W_2$ implies compactness under $\mathrm{SW}_2$. A simple example satisfying Assumption 1 is given by $\Theta = \left\{\nu \in \mathcal{P}_2(\mathbb{R}^d) : \mathrm{supp}(\nu) \subset B(0,R)\right\}$, for some fixed $R < \infty$, where $B(0,R)$ denotes the Euclidean ball of radius $R$ centered at the origin. To verify that this example satisfies Assumption 1, note that since every $\nu \in \Theta$ is supported on the closed ball $B(0,R)$, uniform tightness holds trivially by taking $\mathcal{K} = B(0,R)$, which gives $\nu(\mathcal{K}) = 1 \geq 1 - \varepsilon$ for all $\nu \in \Theta$ and all $\varepsilon > 0$. Uniform integrability of second moments also holds trivially: for any $r > R$, the condition $\|x\| \geq r$ is incompatible with $x \in B(0,R)$, so

$$\sup_{\nu \in \Theta} \int_{\|x\| \geq r} \|x\|^2 \mathrm{d}\nu(x) = 0,$$

and the limit as $r \to \infty$ is therefore zero.

With this assumption, we now show that dDPP with a SW kernel is a well-defined point process.

**Theorem 1** (Well-definedness)**.** *Under Assumption 1 on $\Theta$, distributional determinantal point process with sliced Wasserstein kernel (2) as the L-ensemble kernel is a well-defined DPP on $\Theta$.*

The proof of Theorem 1 is given in Section A.1 of the Supplementary Materials. For the symmetric positive definiteness of the SW kernel, we provide a direct proof rather than relying on the conditional negative definiteness of SW distance as in existing works (Kolouri et al. 2016). Let $T_L : \mathbb{L}_2(\Theta, P_0) \to \mathbb{L}_2(\Theta, P_0)$ be the integral operator of $L$ (compare (12) in the proof):

$$(T_L f)(\nu) = \int_\Theta L(\nu, \nu') f(\nu') \mathrm{d}P_0(\nu').$$

We prove that $\sum_{k=0}^{\infty} \frac{1}{k!} \int_{\Theta^k} \det \left( L(\nu_i, \nu_j) \right)_{i,j=1}^{k} \mathrm{d}P_0(\nu_1) \ldots \mathrm{d}P_0(\nu_k) := \det(I + T_L)$ is finite by proving $T_L$ is trace-class using Mercer's theorem. While the proof focuses on the SW kernel in (2), it might be adaptable to other SW kernels (Kolouri et al. 2016) and other probability kernels as long as they are symmetric positive definite and the associated integral operator is trace-class.

**Remark 1** (Mercer's theorem)**.** *As shown in the proof of Theorem 1, the SW kernel $L$ is symmetric positive definite and continuous. Under the compactness assumption of $\Theta$ (Assumption 1), Mercer's theorem for general metric spaces (Steinwart & Scovel 2012) applies since $P_0$ is a finite measure on $\Theta$. There exist non-negative eigenvalues $\{\lambda_\ell\}_{\ell\geq 1}$ and an orthonormal basis of eigenfunctions $\{\psi_\ell\}_{\ell\geq 1}$ of $\mathbb{L}_2(\Theta, P_0)$ such that*

$$L(\nu,\nu') = \sum_{\ell=1}^{\infty} \lambda_\ell \psi_\ell(\nu)\psi_\ell(\nu'),$$

*with uniform convergence on $\Theta\times\Theta$.*

With Mercer's theorem, we can define the correlation kernel of the dDPP, which will be later used for posterior inference.

**Definition 2** (Correlation kernel)**.** *Let $T_L$ be the integral operator associated with the SW kernel $L$ on $\mathbb{L}_2(\Theta, P_0)$. Assume that $L$ admits the spectral decomposition $\{(\psi_\ell)\}_{\ell\geq 1}$. The* correlation kernel *$K$ is defined as the integral kernel of the operator*

$$T_K := T_L(I+T_L)^{-1}.$$

*Equivalently, $K$ admits the following expansion:*

$$K(\nu,\nu') = \sum_{\ell=1}^{\infty} \frac{\lambda_\ell}{1+\lambda_\ell}\psi_\ell(\nu)\psi_\ell(\nu'), \qquad \text{for } \nu,\nu'\in\Theta.$$

**Remark 2** (dDPP with the marginal correlation kernel)**.** *Let $K$ denote the correlation operator associated with $L$ and $P_0$. For every $k\geq 1$, the $k$-th order correlation function (or $k$-point intensity function) is defined as*

$$\rho_k(\nu_1,\ldots,\nu_k) = \det\Big(K(\nu_i,\nu_j)\Big)_{i,j=1}^{k}, \qquad \text{for } \nu_1,\ldots,\nu_k\in\Theta.$$

## 3.2 Using Empirical Distributions

We next discuss the approximation quality of the plug-in estimators of the L-ensemble kernel and the correlation kernel. In particular, we derive a concentration bound on the approximation error when we observe i.i.d samples from the underlying distributions. We would like to note that in practice the distributions might be fully observed (e.g., in a discrete form), and therefore plug-in estimation might not be needed.

**Theorem 2.** *Let* $\Theta = \{\nu_1, \ldots, \nu_Q\} \subset \mathcal{P}_2(\mathbb{R}^d)$ *be a finite set of probability measures and* $P_0 = \frac{1}{Q}\sum_{i=1}^{Q} \delta_{\nu_i}$. *For each* $i \in [Q]$, *let* $\hat{\nu}_i^{(m)} = \frac{1}{m}\sum_{t=1}^{m} \delta_{x_t^{(i)}}$ *be the empirical measure formed from* $m$ *i.i.d. samples* $\{x_t^{(i)}\}_{t=1}^{m} \sim \nu_i$, *drawn independently across* $i \in [Q]$. *Suppose* $\operatorname{supp}(\nu_i) \subset B(0,R)$ *for all* $i \in [Q]$ *and some fixed* $R < \infty$. *Define the true and estimated L-ensemble and correlation kernel matrices:*

$$L_{ij} = \exp\left(-\gamma SW_2^2(\nu_i, \nu_j)\right), \qquad \hat{L}_{ij} = \exp\left(-\gamma SW_2^2\left(\hat{\nu}_i^{(m)}, \hat{\nu}_j^{(m)}\right)\right),$$

$$\mathbf{K} = \mathbf{L}(I + \mathbf{L})^{-1}, \qquad \hat{\mathbf{K}} = \hat{\mathbf{L}}(I + \hat{\mathbf{L}})^{-1}.$$

*Let* $\lambda_{\min}$ *denote the smallest eigenvalue of (p.d.)* $L$, *and* $\kappa := (1 + \lambda_{\min}(\mathbf{L}))^{-1} < 1$. *Then for any* $\delta \in (0,1)$, *with probability at least* $1 - \delta$, *the following hold simultaneously:*

$$\|\hat{\mathbf{L}} - \mathbf{L}\|_2 \leq \frac{16\gamma R^2 Q}{\sqrt{m}} \sqrt{\pi + \log(2Q^2/\delta)},$$
$$\|\hat{\mathbf{K}} - \mathbf{K}\|_2 \leq \frac{16\kappa\gamma R^2 Q}{\sqrt{m}} \sqrt{\pi + \log(2Q^2/\delta)}.$$

The proof of Theorem 2 is given in Section A.2 of the Supplementary Materials. The bounds exhibit the parametric $\mathcal{O}(m^{-1/2})$ rate in the number of samples, which is notably dimension-free. The dependence on the number of distributions $Q$ is mild, inside the square root, aside from the linear factor arising from bounding the operator norm of a $Q \times Q$ matrix entrywise. Finally, the correlation kernel enjoys a strictly sharper bound than the L-ensemble kernel by the contraction factor $\kappa < 1$, reflecting the smoothing effect of the map $\mathbf{L} \to \mathbf{L}(I + \mathbf{L})^{-1}$.

**Corollary 1.** *Under the setting of Theorem 2, let $S = \{i_1, \ldots, i_k\} \subset [Q]$ be any index set of size $k \leq Q$, and let $\mathbf{L}_S$ and $\hat{\mathbf{L}}_S$ denote the $k \times k$ principal submatrices of $\mathbf{L}$ and $\hat{\mathbf{L}}$, respectively, and similarly $\mathbf{K}_S$ and $\hat{\mathbf{K}}_S$ for $\mathbf{K}$ and $\hat{\mathbf{K}}$. Then for any $\delta \in (0,1)$, with probability at least $1-\delta$:*

$$\left|\det(\hat{\mathbf{L}}_S) - \det(\mathbf{L}_S)\right| \leq \frac{16\, k(3k)^{k-1}\gamma R^2 Q}{\sqrt{m}} \sqrt{\pi + \log(2Q^2/\delta)}, \tag{4}$$

$$\left|\det(\hat{\mathbf{K}}_S) - \det(\mathbf{K}_S)\right| \leq \frac{16\, k\, 3^{k-1}\kappa\gamma R^2 Q}{\sqrt{m}} \sqrt{\pi + \log(2Q^2/\delta)}, \tag{5}$$

*where $\kappa := (1 + \lambda_{\min}(\mathbf{L}))^{-1} < 1$.*

The proof of Corollary 1 is given in Section A.3 of the Supplementary Materials. This corollary transfers the guarantees of Theorem 2 to the determinants of principal submatrices, which govern the DPP probabilities where $\det(\mathbf{L}_S)$ is proportional to the probability that the dDPP configuration is exactly the subset $S = \{\nu_{i_1}, \ldots, \nu_{i_k}\}$ (i.e., that these and only these atoms are jointly present in a realization of $\Phi$), while $\det(\mathbf{K}_S)$ gives the probability that $S$ is contained in the (random) realized configuration. These probabilities are estimated consistently at the parametric $\mathcal{O}(m^{-1/2})$ rate. The prefactors $(3k)^{k-1}$ and $3^{k-1}$ arise from propagating an entrywise perturbation through a $k \times k$ determinant and are benign for the small subset sizes of interest. The sharper factor in (5) reflects that the entries of $\mathbf{K}$ are bounded by one whereas those of $\mathbf{L}$ are not.

# 4 Clustering Distribution-Valued Data

We observe a dataset $\mathcal{S} = \{F_1, \ldots, F_N\}$ of distributions in $\mathcal{P}_2(\mathbb{R}^d)$, and we aim to infer a random partition of $\mathcal{S}$ or equivalently the latent mixing measure in a mixture model for $F_i$. In this section, we introduce such a mixture model using the discrete dDPP and discuss posterior inference, posterior consistency, and posterior summarization.

While the dDPP is well-defined for general measures, we consider a dDPP mixture model under a simple choice of discrete base space $\Theta$. In particular, we take $\Theta = \{\nu_1, \ldots, \nu_Q\}$ (with $0 < Q < \infty$) and the uniform measure

$$P_0 = \frac{1}{Q} \sum_{i=1}^{Q} \delta_{\nu_i}$$

as the associated base measure. We focus on this finite discrete case for two reasons. First, it admits more tractable posterior inference, yielding a fast practical mixing rate for Markov chain Monte Carlo (MCMC) posterior simulation. Second, it allows us to establish posterior consistency.

Letting $\mathrm{dDPP}(\Theta, P_0)$ be the corresponding dDPP on $\Theta$ with the base measure $P_0$, we propose the following hierarchical generalized Bayesian model:

$$\begin{aligned} \ell(F_i \mid \theta_i) &= \exp\left(-w \mathrm{SW}_2^2(F_i, \theta_i)\right), \\ \theta_i \mid G &\overset{i.i.d}{\sim} G, \quad i = 1, \ldots, N, \end{aligned} \tag{6}$$

where $G$ is a discrete random probability measure whose atoms define the clusters by way of sampling latent distribution-valued variables $\theta_i$ from $G$. Equation (6) links $F_i$ with the clusters. The atoms of the mixing measure $G$ is generated by a DPP, as follows:

$$\begin{aligned} G &= \sum_{h=1}^{m} \frac{s_h}{\sum_{h'=1}^{m} s_{h'}} \delta_{\widetilde{\theta}_h}, \\ \{\widetilde{\theta}_1, \ldots, \widetilde{\theta}_m\} &\sim \mathrm{dDPP}(\Theta, P_0), \\ s_1, \ldots, s_m &\overset{i.i.d}{\sim} \mathrm{Gamma}(\alpha, 1). \end{aligned} \tag{7}$$

where $w > 0$ is a bandwidth parameter controlling the concentration of the likelihood around $\theta_i$ and $G$ is a normalized random measure over $\Theta$. We say $F_i$ belongs to cluster $h$ if $\theta_i = \widetilde{\theta}_h$. Here $\theta_i$ and $\widetilde{\theta}_h$ are themselves distributions (elements of $\Theta$) (adopting conventional notation from the Bayesian nonparametric literature to highlight the hierarchical structure). In summary, we define a random partition of $F_1, \ldots, F_N$ by setting up a mixture model using a generalized Bayes kernel based on SW distance.

Since the random mixing measure $G$ is supported on the finite set $\Theta = \{\nu_1, \dots, \nu_Q\}$, it is fully characterized by the weight it places on each atom. Concretely, identifying

$$\pi_q = G(\{\nu_q\}), \qquad q = 1, \dots, Q,$$

the measure $G$ corresponds one-to-one with a vector $\pi = (\pi_1, \dots, \pi_Q) \in \Delta^{Q-1}$, where

$$\Delta^{Q-1} := \left\{ \pi \in \mathbb{R}^Q : \pi_q \geq 0 \ \forall q, \quad \sum_{q=1}^{Q} \pi_q = 1 \right\}.$$

For later reference we note that (6) and using $\pi$ to represent a mixing measure $G$ implies marginal (general) likelihood $\ell(F_i \mid G) = \sum_q \pi_q \exp\left\{-w\mathrm{SW}_2^2(F, \nu_q)\right\}$. And the posterior on $G$ is characterized by a probability model $\Pi_N(\pi \mid F_1, \dots, F_N)$ as a distribution over $\Delta^{Q-1}$, and we establish the following posterior consistency result.

**Theorem 3** (Posterior Consistency)**.** *Let $G_0 \in \mathcal{P}(\mathcal{P}_2(\mathbb{R}^d))$ be the true data-generating distribution of $F_1, \dots, F_N \overset{i.i.d.}{\sim} G_0$, and assume that*

$$\mathbb{E}_{F \sim G_0}\left[\exp\left(w \sum_{q=1}^{Q} SW_2^2(F, \nu_q)\right)\right] < \infty \tag{8}$$

*(for any $\nu_q \in \Theta$). Then for every $\varepsilon > 0$, we have*

$$\Pi_N\left(\tilde{d}(\pi, \mathcal{M}^*) > \varepsilon \middle| F_1, \dots, F_N\right) \xrightarrow{a.s.} 0 \quad \text{as } N \to \infty,$$

*where $\tilde{d}(\pi, \mathcal{M}^*) = \min_{\pi' \in \mathcal{M}^*} \|\pi - \pi'\|_1$ and $\mathcal{M}^* := \arg\min_{\pi \in \Delta^{Q-1}} R(\pi)$ is the set of minimizers of the population risk $R(\pi) := \mathbb{E}_{F \sim G_0}[-\log \sum_{q=1}^{Q} \pi_q \exp(-w\, SW_2^2(F, \nu_q))]$.*

The proof of Theorem 3 is given in Section A.4 of the Supplementary Materials. As $N \to \infty$, the posterior concentrates on the set $\mathcal{M}^*$ of mixing weights minimizing the population risk $R(\pi)$, and hence the procedure recovers the best mixture representation within $\Theta$.

**Remark 3** (Consistency under empirical approximation)**.** *In the case where each $F_i$ is observed only through an empirical measure $\hat{F}_i^{(m)}$ formed from $m$ i.i.d. samples, the conclusion of Theorem 3 continues to hold (in probability) as $N \to \infty$ and $m = m(N) \to \infty$*

*jointly, with no condition on their relative rates. Indeed, replacing each $F_i$ by $\hat{F}_i^{(m)}$ perturbs the generalized likelihood only through the discrepancies $|SW_2^2(F_i,\nu_q) - SW_2^2(\hat{F}_i^{(m)},\nu_q)|$, which vanish as $m \to \infty$ because empirical measures of a $\mathcal{P}_2(\mathbb{R}^d)$ law converge in $W_2$, hence in $SW_2$ (Nadjahi et al. 2020). Assumption (8) makes these discrepancies uniformly integrable, and hence their average over the $N$ observations tends to* 0*.*

# 5 Posterior Inference

## 5.1 Posterior Characterization

We adapt Theorems 1–3 of Beraha et al. (2025) for the posterior characterization under model (6)-(7), working here with the correlation kernel $K$. While one could sample directly via a reversible-jump scheme operating on the L-ensemble (Xu et al. 2016, Green 1995), we instead adapt the sampler of Beraha et al. (2025), which yields a Pólya-urn-like scheme. Below we write $H(s)$ generically for the prior distribution of the marks $s_h$ in (7), and $h(s)$ for the corresponding density, keeping in mind the example of the Gamma$(\alpha, 1)$ prior. The model augmentation with $U_N$ in the following proposition is introduced in anticipation of a simplification in the conditional posterior for $\bar{G}$. The factor $\{\bar{G}(\Theta)\}^N$ in the gamma normalization constant of $p(U_N \mid G)$ cancels against the $1/\bar{G}(\Theta)$ in the (augmented) joint posterior that arises from $p(\theta_i \mid \bar{G}) = \bar{G}(\theta_i)/\bar{G}(\Theta)$ in (6), greatly simplifying the following posterior conditionals.

**Proposition 1** (Posterior Characterization of dDPP Mixture)**.** *Since $(\Theta, SW_2)$ is a compact space and the dDPP is a well-defined finite point process on $\Theta$ (Theorem 1), the general posterior characterization of Beraha et al. (2025)[Theorem 1 and Example 3] applies directly. Let $\boldsymbol{\theta}^* = (\theta_1^*, \ldots, \theta_k^*)$ denote the distinct occupied atoms among $\theta_1, \ldots, \theta_N$ with counts $n_1, \ldots, n_k$, and let $U_N$ be an auxiliary variable with $U_N \mid G \sim$* Gamma$(N, \bar{G}(\Theta))$*,* *where* $\bar{G} = \sum_{h=1}^{k} s_h \, \delta_{\theta_h^*}$ *is the unnormalized random measure. The posterior of* $\bar{G}$ *given* $\boldsymbol{\theta}^*$ *and* $U_N = u$ *equals in distribution:*

$$(\bar{G} \mid \boldsymbol{\theta}^*, U_N = u) \stackrel{d}{=} \sum_{j=1}^{k} S_j^* \, \delta_{\theta_j^*} + \bar{G}',$$

*where* (i) $S_1^*, \ldots, S_k^*$ *are independent with densities* $f_{S_j^*}(s) \propto e^{-us} s^{n_j} h(s)$*,* (ii) $\bar{G}'$ *is an independent residual random measure whose unmarked point process is a dDPP on* $\Theta$ *with correlation kernel*

$$K'(\theta_{\text{new}}, \theta') = K(\theta_{\text{new}}, \theta') - \sum_{i,j=1}^{k} (K_{\boldsymbol{\theta}^*}^{-1})_{ij} \, K(\theta_{\text{new}}, \theta_i^*) \, K(\theta', \theta_j^*),$$

*where* $K_{\boldsymbol{\theta}^*}$ *is the* $k \times k$ *matrix with entries* $K(\theta_i^*, \theta_j^*)$*, and marks drawn i.i.d. from* $f_{S'}(s; u) \propto e^{-su} h(s)$*, and* (iii) *the conditional density of* $U_N$ *given* $\boldsymbol{\theta}^*$ *satisfies*

$$f_{U_N | \boldsymbol{\theta}^*}(u) \propto u^{N-1} \prod_{j \geq 1} \left(1 - \lambda_j^{\boldsymbol{\theta}^*} + \lambda_j^{\boldsymbol{\theta}^*} \, \psi(u)\right) \prod_{j=1}^{k} \kappa(u, n_j),$$

*where* $\{\lambda_j^{\boldsymbol{\theta}^*}\}_{j \geq 1}$ *are the eigenvalues of* $K'$*,* $\psi(u) = \int_{\mathbb{R}_+} e^{-us} h(s) \, ds$ *is the Laplace transform of the mark distribution* $H$*, and* $\kappa(u, n) = \int_{\mathbb{R}_+} e^{-us} s^n h(s) \, ds$*, with closed form expressions under the Gamma*$(\alpha, 1)$ *prior in* (7)*. The residual kernel* $K'$ *is the Schur complement of* $K_{\boldsymbol{\theta}^*}$ *in the augmented kernel matrix. In particular, it is small when* $\theta_{\text{new}}$ *is close to the observed atoms* $\theta_j^*$ *in* $SW_2$*, encoding that the posterior residual process avoids regions of* $\Theta$ *already occupied by observed atoms.*

**Proposition 2** (Marginal and Predictive Distributions of dDPP Mixture)**.** *Under the same conditions as Proposition 1, the marginal distribution of* $\theta_1, \ldots, \theta_N$ *with* $k$ *distinct values* $\boldsymbol{\theta}^*$ *and counts* $n_1, \ldots, n_k$ *(Beraha et al. 2025)[Theorem 2 and Example 5] is:*

$$P(\theta_1, \ldots, \theta_N \in d\boldsymbol{\theta}) = \int_{\mathbb{R}_+} \frac{u^{N-1}}{\Gamma(N)} \prod_{j \geq 1} \left(1 - \lambda_j^{\boldsymbol{\theta}^*} + \lambda_j^{\boldsymbol{\theta}^*} \psi(u)\right) \prod_{j=1}^{k} \kappa(u, n_j) \, du$$
$$\times \det\Big(K(\theta_i^*, \theta_j^*)\Big)_{i,j=1}^{k} \prod_{j=1}^{k} P_0(d\theta_j^*).$$

*The determinant factor* $\det(K(\theta_i^*, \theta_j^*))_{i,j=1}^{k}$ *is the dDPP factorial moment measure density with respect to* $P_0^k$*, which is large when the distinct atoms* $\theta_1^*, \ldots, \theta_k^*$ *are well-separated under*

*$SW_2$, directly reflecting the repulsive prior. Conditionally on $\boldsymbol{\theta}^*$ and $U_N = u$, the predictive distribution of $\theta_{N+1}$ (Beraha et al. 2025)[Theorem 3 and Example 7] is:*

$$P(\theta_{N+1} \in \mathrm{d}\theta_{\mathrm{new}} \mid \boldsymbol{\theta}^*, U_N = u) \propto \sum_{j=1}^{k} \frac{\kappa(u, n_j + 1)}{\kappa(u, n_j)} \delta_{\theta_j^*}(d\theta_{\mathrm{new}})$$

$$+ \kappa(u,1) \frac{\prod_{j \geq 1}(1 - \lambda_j^{\boldsymbol{\theta}^*, \theta_{\mathrm{new}}} + \lambda_j^{\boldsymbol{\theta}^*, \theta_{\mathrm{new}}} \psi(u))}{\prod_{j \geq 1}(1 - \lambda_j^{\boldsymbol{\theta}^*} + \lambda_j^{\boldsymbol{\theta}^*} \psi(u))} \Delta_K(\boldsymbol{\theta}^*, \theta_{\mathrm{new}}) P_0(\mathrm{d}\theta_{\mathrm{new}}), \tag{9}$$

*where $\lambda_j^{\boldsymbol{\theta}^*, \theta_{\mathrm{new}}}$ are the eigenvalues of $K'$ evaluated at the augmented atom set $(\boldsymbol{\theta}^*, \theta_{\mathrm{new}})$, and*

$$\Delta_K(\boldsymbol{\theta}^*, \theta_{\mathrm{new}}) = K(\theta_{\mathrm{new}}, \theta_{\mathrm{new}}) - K_{\boldsymbol{\theta}^*, \theta_{\mathrm{new}}}^{\top} K_{\boldsymbol{\theta}^*}^{-1} K_{\boldsymbol{\theta}^*, \theta_{\mathrm{new}}}, \tag{10}$$

*with $K_{\boldsymbol{\theta}^*, \theta_{\mathrm{new}}} = (K(\theta_{\mathrm{new}}, \theta_1^*), \ldots, K(\theta_{\mathrm{new}}, \theta_k^*))^{\top}$, is the SW repulsion term. Since $SW_2(\theta_{\mathrm{new}}, \theta_{\mathrm{new}}) = 0$, we have $\Delta_K(\boldsymbol{\theta}^*, \theta_{\mathrm{new}}) \in [0,1]$, it approaches 0 as $\theta_{\mathrm{new}} \to \theta_j^*$ for any $j$ (the new atom is $SW_2$-close to an existing one) and is maximized when $\theta_{\mathrm{new}}$ is $SW_2$-distant from all current atoms (maximal distributional diversity). This gives a precise geometric meaning to repulsion in $\mathcal{P}_2(\mathbb{R}^d)$ i.e., new mixture components are penalized proportionally to their sliced Wasserstein proximity to existing ones.*

## 5.2 Posterior Simulation

We adapt the marginal MCMC sampler of Beraha et al. (2025) to the discrete dDPP mixture. Since $\Theta = \{\nu_1, \ldots, \nu_Q\}$ is finite, it becomes straightforward to use (10) to analytically integrate out the random measure $G$. Posterior inference proceeds over the latent assignments $\mathbf{c} = (c_1, \ldots, c_N)$ with $c_i \in \{1, \ldots, Q\}$ such that $\theta_i = \nu_{c_i}$, together with the auxiliary variable $U_N$. Throughout, we write $\boldsymbol{\theta}^* = (\theta_1^*, \ldots, \theta_k^*)$ for the distinct occupied atoms among $\{\theta_j\}_{j=1}^N$, $n_j$ for the count of observations assigned to $\theta_j^*$, $K_{\boldsymbol{\theta}^*}$ for the $k \times k$ submatrix of $\mathbf{K}$ with entries $K(\theta_i^*, \theta_j^*)$, and $K_{\boldsymbol{\theta}^*, \theta_{\mathrm{new}}} = (K(\theta_{\mathrm{new}}, \theta_1^*), \ldots, K(\theta_{\mathrm{new}}, \theta_k^*))^{\top}$ for the vector of kernel evaluations between a candidate atom $\theta_{\mathrm{new}}$ and the currently occupied atoms. The algorithm iterates over three steps.

**Update $U_N$.** Given the current assignments $\mathbf{c}$ with $k$ distinct atoms $\boldsymbol{\theta}^*$, we sample $U_N$ via Metropolis-Hastings targeting the density in Proposition 1(iii):

$$f_{U_N|\boldsymbol{\theta}^*}(u) \propto u^{N-1} \prod_{j\geq 1} \left(1 - \lambda_j^{\boldsymbol{\theta}^*} + \lambda_j^{\boldsymbol{\theta}^*}\,\psi(u)\right) \prod_{j=1}^{k} \kappa(u, n_j),$$

where $\{\lambda_j^{\boldsymbol{\theta}^*}\}_{j\geq 1}$ are the eigenvalues of the residual correlation kernel $K'$ defined in Proposition 1, $\psi(u) = (1+u)^{-a}$, and $\kappa(u,n) = \Gamma(n+a)/\Gamma(a)(1+u)^{-(n+a)}$ for $\mathrm{Ga}(a,1)$ marks. We use a log-normal Metropolis-Hastings proposal $U_N' = U_N \exp(\epsilon)$ with $\epsilon \sim \mathcal{N}(0,\sigma^2)$.

**Update assignments.** For $i = 1,\ldots,N$, we remove observation $i$ from the current configuration and let $\boldsymbol{\theta}^*_{(-i)}$ denote the remaining distinct atoms with counts $n_j^{(-i)}$. We then sample the new assignment $c_i$ from a categorical distribution with unnormalized log-weights:

$$\log p_q \propto \begin{cases} \log \dfrac{\kappa(u,\, n_q^{(-i)}+1)}{\kappa(u,\, n_q^{(-i)})} + \log \ell(F_i \mid \nu_q) & \text{if } \nu_q \in \boldsymbol{\theta}^*_{(-i)}, \\[2ex] \log \kappa(u,1) + \log \widetilde{r}\left(\boldsymbol{\theta}^*_{(-i)}, \nu_q; u\right) + \log \ell(F_i \mid \nu_q) - \log n_{\text{aux}} & \text{if } \nu_q \notin \boldsymbol{\theta}^*_{(-i)}, \end{cases}$$

where $\ell(F_i \mid \nu_q) = \exp(-w\,\mathrm{SW}_2^2(F_i, \nu_q))$ is the SW generalized likelihood and

$$\log \widetilde{r}\left(\boldsymbol{\theta}^*_{(-i)}, \nu_q; u\right) \approx \left(\mathrm{Tr}(K'_{(\boldsymbol{\theta}^*_{(-i)},\nu_q)}) - \mathrm{Tr}(K'_{\boldsymbol{\theta}^*_{(-i)}})\right)(\psi(u) - 1)$$

is the Le Cam approximation (Beraha et al. 2025) to the log-ratio of Laplace functionals in Proposition 2 (second line of (9)), with $\mathrm{Tr}(K'_{\boldsymbol{\theta}^*_{(-i)}})$ the trace of the residual kernel after removing observation $i$ and $\mathrm{Tr}(K'_{(\boldsymbol{\theta}^*_{(-i)},\nu_q)})$ the trace after further adding $\nu_q$. For new atoms, we propose $n_{\text{aux}}$ (is later set to 1) candidates drawn proportionally to $\Delta_K(\boldsymbol{\theta}^*_{(-i)}, \cdot)$ from the unoccupied atoms in $\Theta$, following the auxiliary variable scheme of Neal (2000).

**Update occupied atoms.** Given the current assignments $\mathbf{c}$ and occupied atoms $\boldsymbol{\theta}^*$, we update each occupied atom $\theta_h^*$ via a Metropolis-Hastings step. We propose swapping $\theta_h^*$ with a uniformly drawn unoccupied atom $\theta_{\text{prop}} \in \Theta \setminus \boldsymbol{\theta}^*$ and accept using log acceptance ratio:

$$\log \alpha = \left(\log \det K_{\boldsymbol{\theta}^*_{\text{prop}}} - \log \det K_{\boldsymbol{\theta}^*}\right) + \sum_{\{i:\, c_i = h\}} \left(\log \ell(F_i \mid \theta_{\text{prop}}) - \log \ell(F_i \mid \theta_h^*)\right),$$

where $\boldsymbol{\theta}^*_{\text{prop}}$ is $\boldsymbol{\theta}^*$ with $\theta^*_h$ replaced by $\theta_{\text{prop}}$. The first term favors configurations where the occupied atoms are well-separated under $\text{SW}_2$ via the dDPP prior, and the second term arises from the SW likelihood of the observations assigned to cluster $h$.

**Remark 4.** *The $Q \times Q$ kernel matrices $\mathbf{L}$ and $\mathbf{K}$ are precomputed once from the pairwise $SW_2$ distances among $\nu_1, \ldots, \nu_Q$, costing $\mathcal{O}(Q^2)$ kernel evaluations. The SW repulsion term $\Delta_K(\boldsymbol{\theta}^*_{(-i)}, \nu_q)$ is computed via a rank-one Cholesky update of $K_{\boldsymbol{\theta}^*_{(-i)}}$, costing $\mathcal{O}(k^2)$ per candidate atom rather than $\mathcal{O}(k^3)$ from scratch. The SW generalized likelihoods $\ell(F_i \mid \nu_q)$ for all $q = 1, \ldots, Q$ are also precomputed from the pairwise $SW_2$ distances between observations $F_i$ and atoms $\nu_q$, which are fixed throughout the MCMC.*

## 5.3 Posterior Summarization

For posterior summarization, we extend the approach of Nguyen & Mueller (2026) to report a point estimate of the mixing measure $G$, which in turn implies the random partition. Although the marginal MCMC integrates out $G$ analytically and targets the posterior over the assignments $\mathbf{c} = (c_1, \ldots, c_N)$ and the auxiliary variable $U_N$, samples of the mixing measure itself are recovered at no additional cost by exploiting the conditional in Proposition 1. At the end of each iteration, the sampler yields $k$ distinct occupied atoms $\boldsymbol{\theta}^* = (\theta^*_1, \ldots, \theta^*_k)$, counts $n_1, \ldots, n_k$, and $U_N = u$; the unnormalized weights $S^*_1, \ldots, S^*_k$ are then conditionally independent with densities $f_{S^*_j}(s) \propto e^{-us}\, s^{n_j} h(s)$, which for $H = \text{Gamma}(a, 1)$ reduces to $S^*_j \mid \boldsymbol{\theta}^*, U_N = u \sim \text{Gamma}(n_j + a, 1 + u)$ independently across $j$. Normalizing yields $\pi_j = s_j / \sum_{j'} s_{j'}$. We record the mixing measure as $G = \sum_{j=1}^k \pi_j\, \delta_{\theta^*_j}$. The full posterior also includes a residual dDPP component on $\Theta \setminus \boldsymbol{\theta}^*$ (Proposition 1), but since residual atoms carry no observational support, their posterior weights are negligible and this component is omitted in practice. After obtaining a point estimate of the mixing measure $\hat{G} := \sum_{j=1}^k \pi_j \delta_{\theta^*_j}$, a point estimate of the cluster label $c$ for any data point $F$ can be obtained using maximum conditional probabilities:

$$c = \arg\max_{j \in \{1,\dots,k\}} \log p(c = j \mid \hat{G}, F) = \arg\max_{j \in \{1,\dots,k\}} \log \pi_j - w\,\mathrm{SW}_2^2(F, \theta_j^*).$$

The key challenge of the default proposed in Nguyen & Mueller (2026) is that $G$ is a measure over distributions, which calls for a different utility function. Let $W_{\mathrm{SW}_2^2}(\hat{G}, G)$ denote Wasserstein distance with $\mathrm{SW}_2^2$ as the ground cost. We report a point estimate of $G$ based on the loss $\ell(\hat{G}, G) = W_{\mathrm{SW}_2^2}(\hat{G}, G)$, and therefore posterior expected loss

$$\mathbb{L}(\hat{G}) = \mathbb{E}\left[ W_{\mathrm{SW}_2^2}\Big(\hat{G}, G\Big) \,\Big|\, F_1, \dots, F_N \right].$$

We then report a point estimate $\hat{G}$ as the Bayes rule by minimizing posterior expected loss $\mathbb{L}(\hat{G})$. Using imputed posterior MCMC samples $G_t$ across iterations $t = 1, \dots, T$, we approximate $\mathbb{L}$ as a Monte Carlo average, reporting the point estimate:

$$\hat{G} = \arg\min_{G \in \{G_1, \dots, G_T\}} \frac{1}{T} \sum_{t=1}^{T} W_{\mathrm{SW}_2^2}\Big(G, G_t\Big). \tag{11}$$

The resulting point estimate $\hat{G}$ is thus selected among the MCMC samples $\{G_1, \dots, G_T\}$ as the Bayes rule (11) minimizing the average $W_{\mathrm{SW}_2^2}$ distance to all other samples. Selecting the estimate requires the pairwise distance matrix among the $T$ samples, giving $\mathcal{O}(T^2)$ evaluations of $W_{\mathrm{SW}_2^2}$. Each evaluation in turn solves an optimal transport problem over the at most $k$ mixture components, whose ground-cost entries are SW distances between distributions with at most $m$ atoms. The overall cost is therefore $\mathcal{O}\Big(T^2(k^3 \log k + k^2 m \log m)\Big)$, where the $k^3 \log k$ term is the optimal transport solve and $k^2 m \log m$ accounts for forming the $\mathrm{SW}_2^2$ ground-cost matrix.

# 6 Examples

In the following two applications, we used the D2PP for inference on random partitions of distribution-valued data. We compare against inference under a Ferguson-Dirichlet process

(DP) prior for $G$ using the same base measure. Both are used as priors in the generalized Bayes framework of (6)-(7). For the DP, we implement Neal's Algorithm 8 (Neal 2000), augmented with an update step for occupied atoms analogous to that used for the dDPP. Since the models differ only in their choice of prior, the comparison is controlled and fair. We conduct experiments on population-scale single-cell dataset OneK1K (Yazar et al. 2022) in Section 6.1 and on data from the Human Epilepsy Project (HEP) (French et al. 2012) data in Section 6.2. In both cases, we use the set of data points $\mathcal{S}$ as the atom set $\Theta$ with the base measure $P_0$ being a uniform distribution over $\Theta$, for both DP and dDPP. For the generalized-likelihood hyperparameter $w$, we consider values chosen relative to the scale of the pairwise $\mathrm{SW}_2^2$ distances between data points in $\mathcal{S}$, using the inverse of their median as a reference scale. For evaluation, we report the posterior expectations of the number of clusters, mean cluster size, log generalized likelihood, and repulsion (measured by the log unnormalized density of the dDPP prior). We also compute these quantities for the point estimates obtained via the Bayes rule (11). In all experiments, we use 1000 projections to approximate SW distance and SW kernels. Those projections are kept fixed during posterior inference without resampling. We run 2000 MCMC iterations with 1000 iterations for burn-in.

## 6.1 Single-Cell Data

Population-scale single-cell RNA-sequencing count matrices (with thousands of cells and 10,786 genes) were obtained for four immune cell populations (B cells, T cells, natural killer (NK) cells, and monocytes) for each donor. Donors with available data across all four cell types were retained, yielding a common cohort of 961 donors. For each donor, cells from all four cell types were pooled and preprocessed using per-cell total-count normalization followed by a $\log(1+x)$ transformation $\tilde{x}_{ij} = \log\left(1 + \frac{x_{ij}}{\sum_{j'} x_{ij'}}\right)$, where $x_{ij}$ denotes the raw count of gene $j$ in cell $i$. A donor-level mean expression matrix $M \in \mathbb{R}^{961\times 10{,}786}$ was constructed

Table 1: Single cell data. The table reports inference for the number of clusters $k$, average cluster size, log generalized-likelihood, and repulsion (log unnormalized dDPP density): $\log\det(\exp(-\mathrm{SW}_2^2(\cdot,\cdot))))$. "Point" is the Bayes rule (11); $\mathbb{E}$ and HCI refer to posterior expectation and the 95% highest-credible interval (HCI) based on the last $T$ samples.

| | | Clusters $k$ | | Average Cluster Size | | Log-generalized likelihood | | Repulsion | |
|---|---|---|---|---|---|---|---|---|---|
| **Model** | $w$ | **Point** | $\mathbb{E}$ and [HCI] | **Point** | $\mathbb{E}$ and [HCI] | **Point** | $\mathbb{E}$ and [HCI] | **Point** | $\mathbb{E}$ and [HCI] |
| DP | $5\times10^5$ | 14 | 14.45 [11.00, 17.00] | 68.64 | 67.46 [50.58, 80.08] | −4.17 | −4.17 [−4.18, −4.16] | −183.58 | −196.71 [−254.93, −146.09] |
| | $7\times10^5$ | 24 | 23.86 [21.00, 27.00] | 40.04 | 40.51 [35.59, 45.76] | −5.16 | −5.16 [−5.18, −5.15] | −341.23 | −361.61 [−430.48, −300.96] |
| | $10^6$ | 37 | 34.77 [31.00, 38.00] | 25.97 | 27.73 [25.29, 31.00] | −6.35 | −6.36 [−6.38, −6.34] | −612.27 | −575.06 [−639.70, −509.13] |
| dDPP ($\gamma=10^3$) | $5\times10^5$ | 11 | 11.77 [11.00, 13.00] | 87.36 | 82.06 [73.92, 87.36] | −4.21 | −4.22 [−4.23, −4.21] | −133.79 | −149.21 [−172.34, −132.02] |
| | $7\times10^5$ | 21 | 22.23 [20.00, 24.00] | 45.76 | 43.34 [38.44, 45.76] | −5.26 | −5.26 [−5.27, −5.24] | −313.32 | −336.60 [−364.19, −300.58] |
| | $10^6$ | 32 | 33.04 [31.00, 36.00] | 30.03 | 29.15 [26.69, 31.00] | −6.42 | −6.45 [−6.49, −6.42] | −523.72 | −542.00 [−597.89, −508.47] |
| dDPP ($\gamma=10^4$) | $5\times10^5$ | 11 | 11.68 [11.00, 13.00] | 87.36 | 82.48 [73.92, 87.36] | −4.24 | −4.25 [−4.25, −4.24] | −134.30 | −146.04 [−170.94, −131.94] |
| | $7\times10^5$ | 20 | 19.76 [18.00, 22.00] | 48.05 | 48.83 [43.68, 53.39] | −5.23 | −5.25 [−5.28, −5.23] | −287.61 | −285.09 [−325.09, −261.32] |
| | $10^6$ | 33 | 33.08 [30.00, 36.00] | 29.12 | 29.14 [26.69, 32.03] | −6.49 | −6.49 [−6.52, −6.47] | −519.52 | −527.78 [−585.97, −470.31] |
| dDPP ($\gamma=10^5$) | $5\times10^5$ | 12 | 11.88 [10.00, 13.00] | 80.08 | 81.35 [73.92, 96.10] | −4.19 | −4.20 [−4.22, −4.19] | −158.23 | −152.19 [−173.01, −120.30] |
| | $7\times10^5$ | 20 | 21.79 [20.00, 25.00] | 48.05 | 44.31 [38.44, 48.05] | −5.26 | −5.26 [−5.29, −5.24] | −301.51 | −327.03 [−379.44, −287.03] |
| | $10^6$ | 34 | 35.98 [32.00, 39.00] | 28.26 | 26.79 [24.03, 29.12] | −6.45 | −6.47 [−6.53, −6.43] | −558.91 | −594.74 [−650.37, −518.95] |

by averaging the normalized, log-transformed expression profiles of all cells belonging to each donor. We applied principal component analysis (PCA) to $M$ and retained the top 17 principal components (cumulative explained variance about 95%). The fitted PCA projection was then applied to every individual cell of each donor, producing for each donor $p$ an empirical distribution of cells in the 17-dimensional PCA space.

The results in Table 1 report more parsimonious and better-separated partitions for inference under the dDPP than under the DP baseline. Across a range of values for the likelihood hyperparameter $w$ dDPP reports fewer clusters, with correspondingly larger average cluster sizes. Repulsion is *a priori* determined by the repulsion hyperparameter $\gamma$ of the SW kernel. Smaller values of $\gamma$ induce stronger prior repulsion between atoms, which typically yields fewer, more separated clusters. However, practically, inference on the number of clusters remains robust with respect to choice of $\gamma$ in a wide range, confirming appropriate

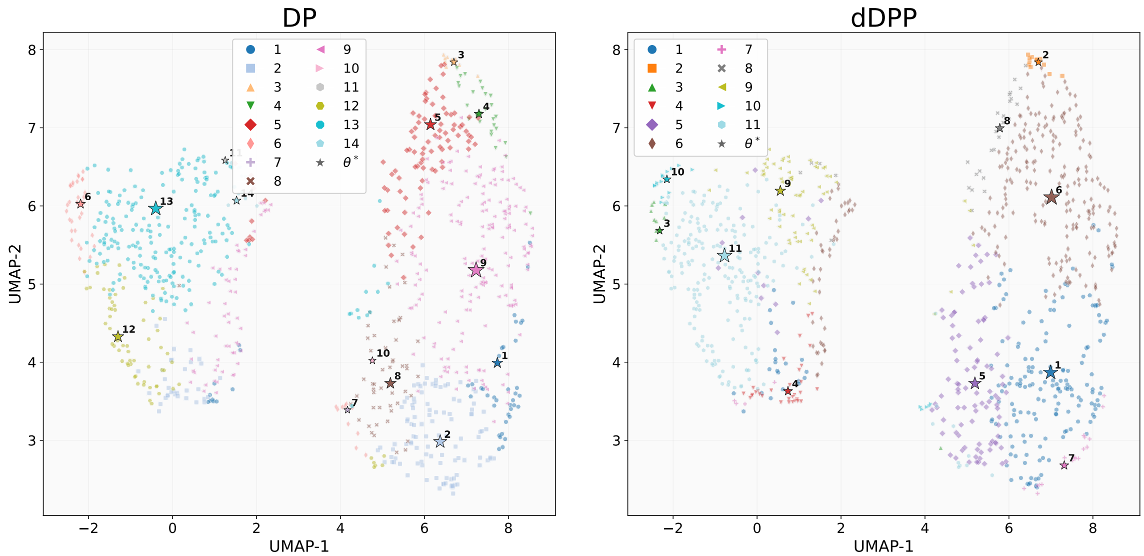



Figure 1: Single cell data. UMAP visualization based on point estimates 11 of the mixing measures under the DP and dDPP models ($w = 5 \times 10^5, \gamma = 10^3$). Colors indicate the corresponding partitions. Atoms of the mixing measures are shown as stars, with size proportional to their weights.

dominance of the likelihood. Similar behavior appears in the repulsion column in Table 1. Inference under the dDPP attains substantially higher (less negative) log unnormalized density at every weight, confirming that its atoms occupy more distinct regions of the embedding. Importantly, this improved separation of clusters costs almost nothing in log generalized likelihood. Inference under the dDPP trades a negligible amount of fit for a markedly more separated and interpretable clustering. It also produces more concentrated posteriors over partition complexity, with 95% highest-credible intervals (HCIs) as tight as or tighter than the DP's.

Figures 1 and 2 show the point estimate (11) under $w = 5 \times 10^5$ and $\gamma = 10^3$. In the UMAP comparison of Figure 1, the dDPP solution (right) assigns donors to fewer, spatially more coherent groups, with atoms better spread out across diverse regions of the embedding. In contrast, the estimate under the DP model (left) clusters the atoms more tightly and

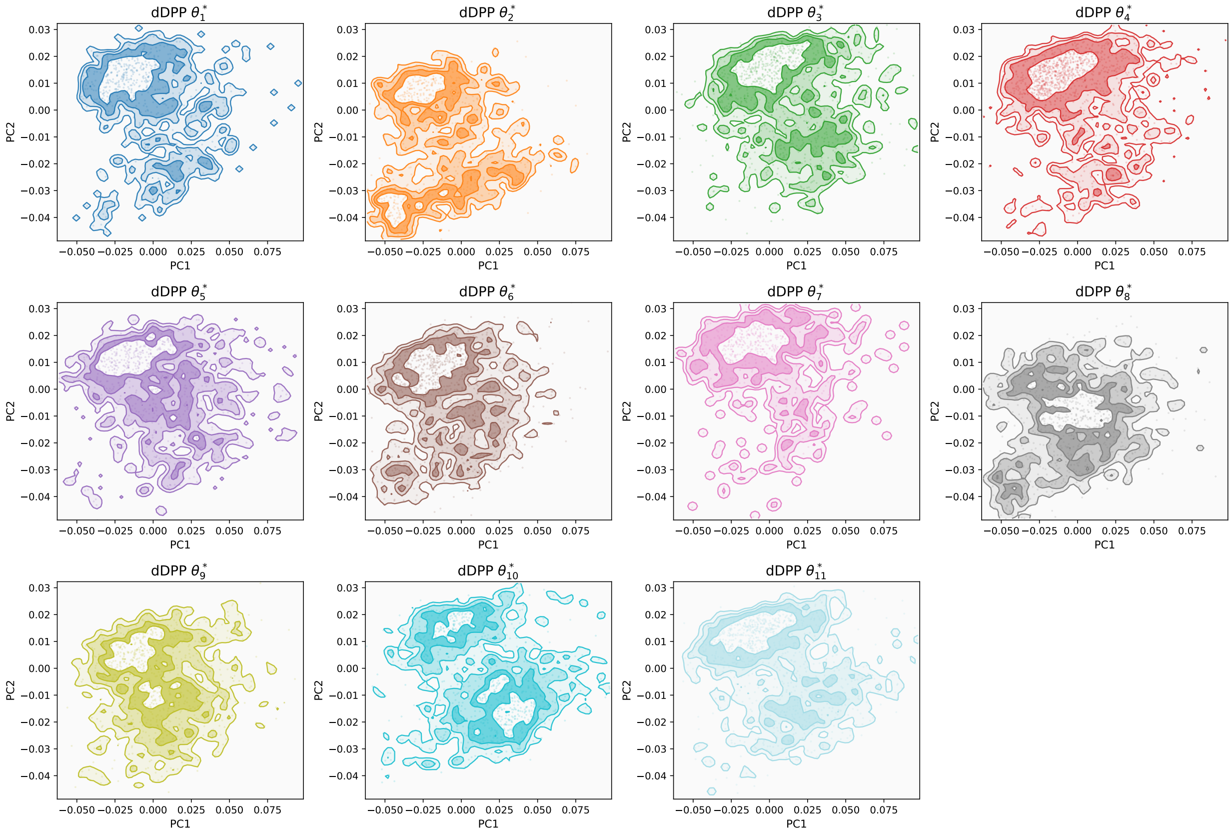



Figure 2: Singe cell data - PCA visualization of the atoms of the mixing measure. The figure shows kernel density estimates of $\hat{G}$ under (11) ($w = 5 \times 10^5, \gamma = 10^3$).

yields a more fragmented partition. Analogous figures for $w = 7 \times 10^5$ and $w = 10^6$ appear in Figure 6 of Section B.1 of the Supplementary Materials, and PCA visualizations of the donor clusters for $w = 5 \times 10^5$ are shown in Figures 7–10. In each case, the donors within a cluster closely resemble the typical donors $\theta_k^*$ shown in Figure 2. Figure 2 overlays the dDPP atoms over a kernel density estimate of the mixture model corresponding to $\hat{G}$ in (11). The figure illustrates how the recovered atoms align with the dominant modes of the observed data. The corresponding figure for the DP atoms is given in Figure 11 of Section B.1 of the Supplementary Materials.

## 6.2 Human Epilepsy Project Data

The Human Epilepsy Project (French et al. 2012) (HEP) enrolled 448 individuals newly diagnosed with focal epilepsy between 2012 and 2017 across 34 clinical centers worldwide. Eligible participants were between ages 12 and 60 at diagnosis and enrolled within four months of initiating medical treatment. Participants recorded daily seizure counts via an electronic seizure diary over a three-year follow-up period. After excluding individuals with no diary entries, no properly tracked days, or no medication data, the final analysis cohort consisted of 407 individuals. For the illustrative purpose of this example, we only cluster the seizure data (not using treatment and baseline covariates). For participant $i$, the daily record reports $x_{i,t} \in \{0, 0.5, 1\}$ for $t = 1, \ldots, T_i$, where 0 = no seizure, 1 = seizure, and 0.5 = missing (imputed). Since $T_i$ varies across participants and patients are enrolled at varying times, the data are unaligned. For meaningful clustering, we map the data for each patient to a distribution of "reads", where reads are defined as length $W$ subsequences as follows. We extract $T_i - W + 1$ consecutive windows of width $W > 1$: $\mathbf{x}_{i,k} = (x_{i,k}, x_{i,k+1}, \ldots, x_{i,k+w-1}) \in \{0, 0.5, 1\}^W, \quad k = 1, \ldots, T_i - W + 1$. Each participant is then represented as an empirical distribution over these reads: $\hat{F}_i = \frac{1}{T_i - w + 1} \sum_{k=1}^{T_i - W + 1} \delta_{\mathbf{x}_{i,k}} \in \mathcal{P}(\{0, 0.5, 1\}^W)$, where $\delta_{\mathbf{x}_{i,k}}$ is a Dirac mass at $\mathbf{x}_{i,k}$ and $\mathcal{P}(\{0, 0.5, 1\}^W)$ denotes the space of probability measures on $\{0, 0.5, 1\}^W$. This places all participants in a common space regardless of follow-up length. We cluster $F_i$, $i = 1, \ldots, N$, using the proposed dDPP model for random partitions with distribution-valued data.

The results on the HEP cohort in Table 2 (with $W = 7$) reproduce the pattern observed on the single-cell data. Across all choices of hyperparameter $w$, dDPP recovers fewer clusters than the DP baseline, with correspondingly larger mean cluster sizes, and the recovered atoms are markedly better separated, as reflected by the substantially higher repulsion score at every weight. The repulsion hyperparameter $\gamma$ governs the strength of this effect.

Table 2: HEP data: Same as Table 1 for HEP data.

| Model | $w$ | Clusters $k$ Point | Clusters $k$ $\mathbb{E}$ and [HCI] | Average Cluster Size Point | Average Cluster Size $\mathbb{E}$ and [HCI] | Log-generalized likelihood Point | Log-generalized likelihood $\mathbb{E}$ and [HCI] | Repulsion Point | Repulsion $\mathbb{E}$ and [HCI] |
|---|---|---|---|---|---|---|---|---|---|
| DP | 100 | 9 | 8.00 [5.00, 11.00] | 45.22 | 53.19 [37.00, 81.40] | −2.12 | −2.11 [−2.13, −2.09] | −52.79 | −45.66 [−82.02, −11.16] |
| | 200 | 14 | 14.06 [11.00, 17.00] | 29.07 | 29.40 [23.94, 37.00] | −2.85 | −2.87 [−2.89, −2.85] | −115.78 | −114.48 [−161.27, −67.52] |
| | 500 | 27 | 26.72 [22.00, 31.00] | 15.07 | 15.35 [12.72, 17.70] | −4.13 | −4.14 [−4.18, −4.10] | −286.84 | −276.72 [−352.94, −216.60] |
| dDPP ($\gamma = 10$) | 100 | 6 | 6.22 [6.00, 7.00] | 67.83 | 65.86 [58.14, 67.83] | −2.10 | −2.10 [−2.12, −2.09] | −13.73 | −18.57 [−41.43, −13.38] |
| | 200 | 12 | 11.87 [11.00, 13.00] | 33.92 | 34.43 [31.31, 37.00] | −2.86 | −2.87 [−2.88, −2.85] | −84.43 | −81.27 [−106.24, −57.26] |
| | 500 | 23 | 23.25 [21.00, 26.00] | 17.70 | 17.56 [15.65, 19.38] | −4.12 | −4.13 [−4.17, −4.10] | −224.18 | −227.01 [−259.23, −189.43] |
| dDPP ($\gamma = 100$) | 100 | 6 | 6.28 [5.00, 8.00] | 67.83 | 65.45 [50.88, 81.40] | −2.10 | −2.11 [−2.13, −2.10] | −13.47 | −17.39 [−43.43, −11.61] |
| | 200 | 13 | 12.23 [11.00, 14.00] | 31.31 | 33.48 [29.07, 37.00] | −2.84 | −2.86 [−2.87, −2.84] | −109.33 | −91.45 [−114.51, −57.46] |
| | 500 | 23 | 24.91 [22.00, 27.00] | 17.70 | 16.39 [15.07, 18.50] | −4.16 | −4.14 [−4.19, −4.11] | −223.72 | −243.78 [−292.54, −216.96] |
| dDPP ($\gamma = 1000$) | 100 | 7 | 7.89 [6.00, 10.00] | 58.14 | 53.18 [40.70, 67.83] | −2.09 | −2.10 [−2.13, −2.09] | −20.24 | −41.56 [−77.90, −13.67] |
| | 200 | 13 | 12.97 [11.00, 15.00] | 31.31 | 31.65 [27.13, 37.00] | −2.84 | −2.86 [−2.88, −2.84] | −88.59 | −100.51 [−143.17, −62.84] |
| | 500 | 26 | 24.49 [21.00, 28.00] | 15.65 | 16.74 [14.54, 19.38] | −4.12 | −4.16 [−4.21, −4.12] | −258.44 | −246.23 [−304.46, −186.18] |

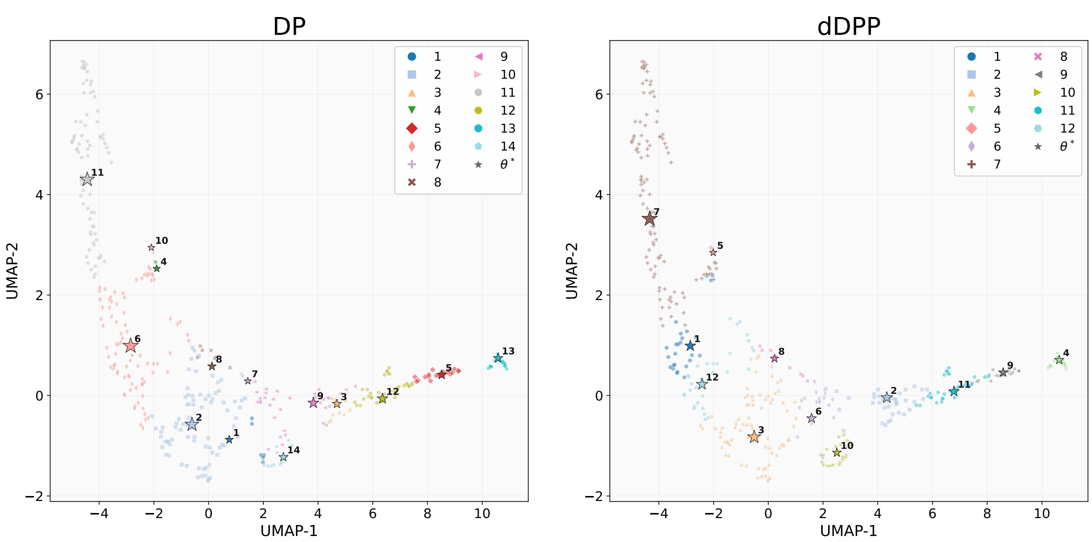



Figure 3: HEP data. Same as Figure 1 with $w = 200, \gamma = 10$, for the HEP data.

This separation again comes at essentially no cost to fit. In particular, the log-likelihood is statistically indistinguishable between the two models throughout. Inference under the dDPP model yields more concentrated posteriors over partition complexity, with 95% HCIs for the number of clusters that are consistently tighter than the DP's. In other words, inference under the dDPP model trades a negligible amount of likelihood for a more

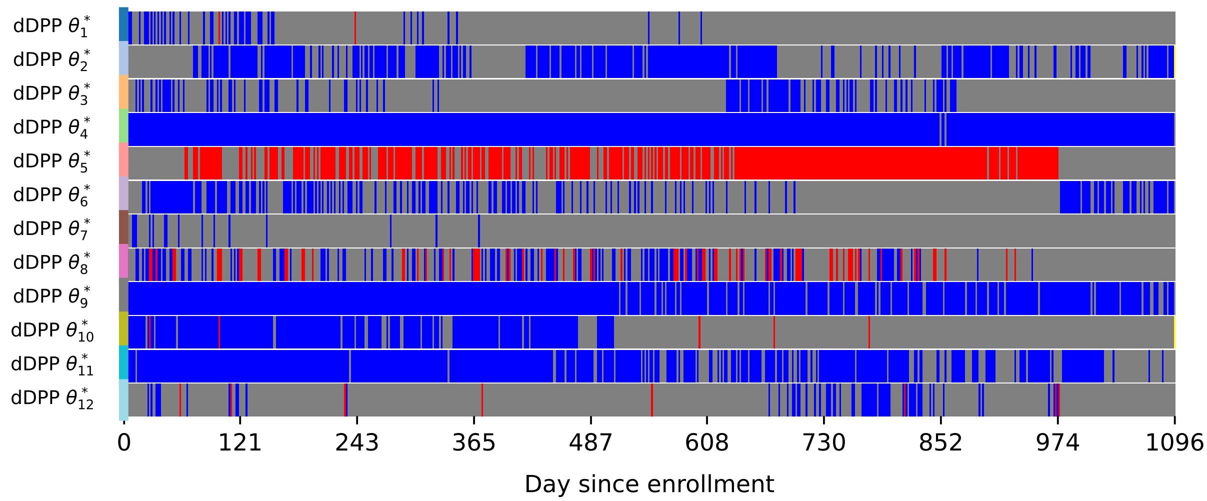



Figure 4: HEP data. The atoms of the point estimate (11) of the mixing measure under the dDPP model ($w = 200, \gamma = 10$). The vertical axis reports the $K = 12$ estimated clusters. The horizontal axis are days of followup. The plot shows $\theta_k^*$. Blue represents 0, red represents 1, and grey represents 0.5; yellow indicates padding where applicable (patients with fewer check-ups).

separated and more interpretable clustering of the seizure-diary distributions.

We again compare the point estimates (11) under the DP and dDPP models. In the UMAP comparison of Figure 3 ($w = 200, \gamma = 10$), the dDPP solution (right) assigns participants to fewer, more spatially coherent groups, with atoms distributed across distinct regions of the embedding, whereas the DP solution (left) clusters its atoms more tightly and yields a more fragmented partition. The same phenomenon appears for $w = 100$ and $w = 500$ in Figure 12 of Section B.2 of the Supplementary Materials. Figure 4 visualizes the recovered dDPP atoms as representative seizure-window patterns, highlighting the distinct temporal profiles each cluster captures; analogous figures for both the DP and dDPP models with $w = 100$, $w = 200, \gamma = 10$, and $w = 500$ are given in Figures 13–15 of Section B.2 of the Supplementary Materials. Finally, Figure 5 displays the full cohort under the induced cluster labels ($w = 200, \gamma = 10$), confirming that the partition organizes participants into

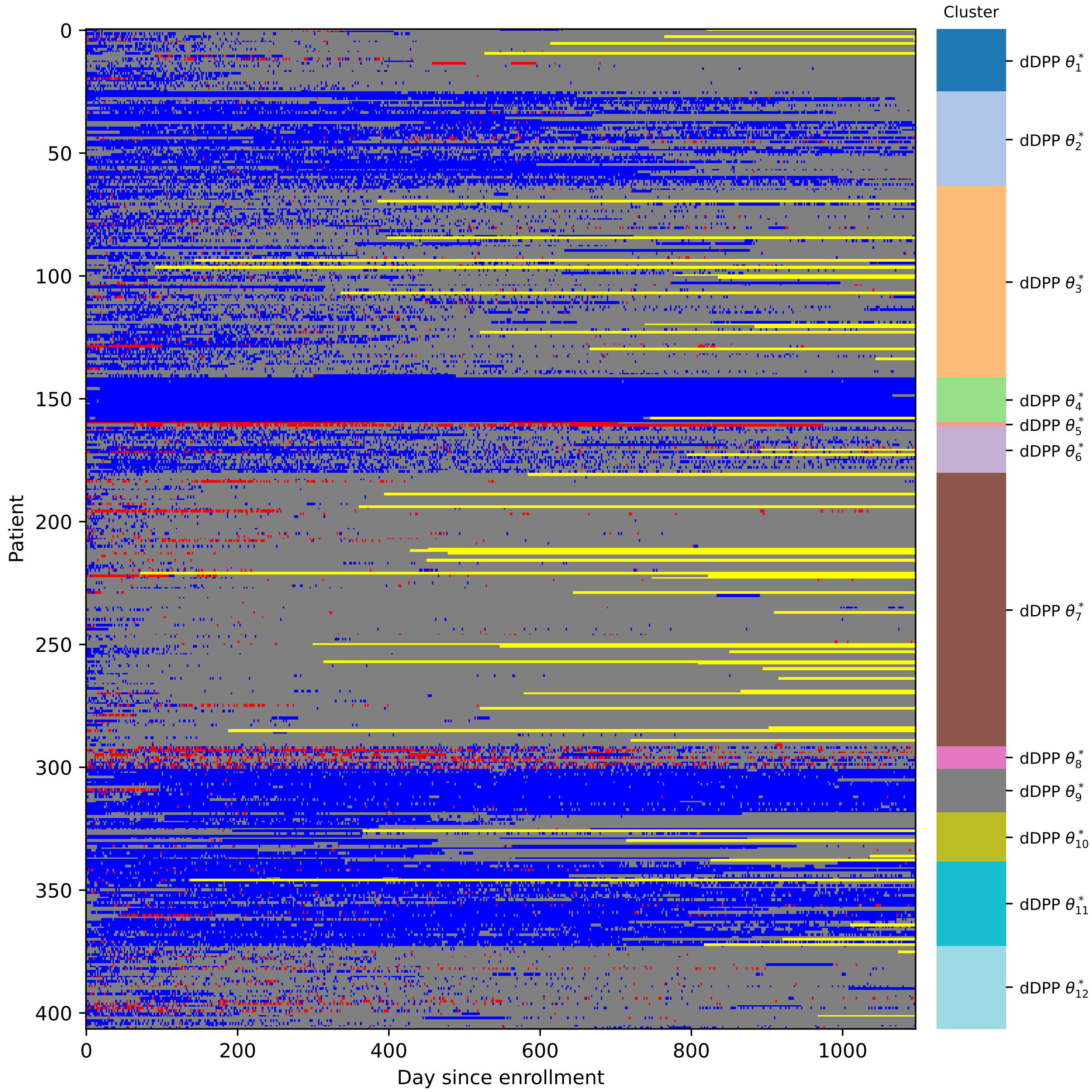



Figure 5: HEP data. Visualization of all patients with corresponding clusters label. Similar to Figure 4, with observed patients $i = 1, \ldots, N$ on the vertical axis, and adding color coding for cluster membership (on the right side) and yellow indicates padding for patients with $T_i < T_{\max}$.

coherent, well-separated groups.

# 7 Conclusion

We introduced the Distributional Determinantal Point Process (dDPP) as a repulsive point process with distribution-valued atoms with Wasserstein kernels to define the L-ensemble kernel. While the use of the generalized likelihood in the hierarchical mixture model construction keeps inference computationally tractable, the same could be seen as compromising the nature of the approach as principled model-based inference. See, for example, McAlinn & Takanashi (2026) for a recent discussion. Another limitation is the use of a finite discrete base measure $P_0$.

Useful generalizations could extend the model to multiple dependent random partitions, as it arises in many applications. The use of common atoms (in this case, distribution-valued atoms) is natural in model (6) and (7), as it already separates the prior on atoms $\widetilde{\theta}_h$ and corresponding marks $s_h$. The latter could vary across dependent random partitions and the earlier could be shared. Moreover, additional theoretical studies of the proposed model can also be investigated such that clustering consistency.

# 8 Data Availability Statement

Single-cell data is available as described in Yazar et al. (2022). Human Epilepsy Project data is available by application (see French et al. (2012)).

# References


Beraha, M., Argiento, R., Camerlenghi, F. & Guglielmi, A. (2025), 'Bayesian mixture models with repulsive and attractive atoms', *Journal of the Royal Statistical Society Series B: Statistical Methodology* **87**(5), 1481–1507.

Beraha, M., Argiento, R., Møller, J. & Guglielmi, A. (2022), 'MCMC computations for Bayesian mixture models using repulsive point processes', *Journal of Computational and Graphical Statistics* **31**(2), 422–435.

Bianchini, I., Guglielmi, A. & Quintana, F. A. (2020), 'Determinantal Point Process Mixtures Via Spectral Density Approach', *Bayesian Analysis* **15**(1), 187 – 214.
**URL:** *https://doi.org/10.1214/19-BA1150*

Bissiri, P. G., Holmes, C. C. & Walker, S. G. (2016), 'A general framework for updating belief distributions', *Journal of the Royal Statistical Society Series B: Statistical Methodology* **78**(5), 1103–1130.

Bonneel, N., Rabin, J., Peyré, G. & Pfister, H. (2015), 'Sliced and Radon Wasserstein barycenters of measures', *Journal of Mathematical Imaging and Vision* **1**(51), 22–45.

Carrière, M., Cuturi, M. & Oudot, S. (2017), Sliced Wasserstein kernel for persistence diagrams, *in* 'Proceedings of the 34th International Conference on Machine Learning-Volume 70', pp. 664–673.

Chakraborty, A., Bhattacharya, A. & Pati, D. (2025), 'Robust probabilistic inference via a constrained transport metric', *Bayesian Analysis* **1**(1), 1–33.

Cremaschi, A., Wertz, T. M. & De Iorio, M. (2025), 'Repulsion, chaos, and equilibrium in mixture models', *Journal of the Royal Statistical Society Series B: Statistical Methodology* **87**(2), 389–432.

Ferguson, T. S. (1973), 'A Bayesian analysis of some nonparametric problems', *The annals of statistics* pp. 209–230.

French, J. A., Kuzniecky, R. & Lowenstein, D. (2012), 'Human epilepsy project', NYU Data Catalog. Dataset, UID 10473; accessed 2026-07-20.
**URL:** *https://datacatalog.med.nyu.edu/dataset/10473*

Green, P. J. (1995), 'Reversible jump Markov chain Monte Carlo computation and Bayesian model determination', *Biometrika* **82**(4), 711–732.

Ho, N., Nguyen, X., Yurochkin, M., Bui, H. H., Huynh, V. & Phung, D. (2017), Multilevel clustering via Wasserstein means, *in* 'International conference on machine learning', PMLR, pp. 1501–1509.

Hough, J. B., Krishnapur, M., Peres, Y. et al. (2009), *Zeros of Gaussian analytic functions and determinantal point processes*, Vol. 51, American Mathematical Soc.

Kingman, J. F. (1978), 'The representation of partition structures', *Journal of the London Mathematical Society* **2**(2), 374–380.

Kolouri, S., Zou, Y. & Rohde, G. K. (2016), Sliced Wasserstein kernels for probability distributions, *in* 'Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition', pp. 5258–5267.

Kulesza, A. & Taskar, B. (2012), 'Determinantal point processes for machine learning', *Foundations and Trends® in Machine Learning* **5**(2-3), 123–286.

Lavancier, F., Møller, J. & Rubak, E. (2015), 'Determinantal point process models and statistical inference', *Journal of the Royal Statistical Society Series B: Statistical Methodology* **77**(4), 853–877.

Leluc, R., Dieuleveut, A., Portier, F., Segers, J. & Zhuman, A. (2024), Sliced-Wasserstein estimation with spherical harmonics as control variates, *in* 'International Conference on Machine Learning', PMLR, pp. 27191–27214.

Lijoi, A., Mena, R. H. & Prünster, I. (2005), 'Hierarchical mixture modeling with normalized inverse-gaussian priors', *Journal of the American Statistical Association* **100**(472), 1278–1291.

Lijoi, A., Mena, R. H. & Prünster, I. (2007), 'Controlling the reinforcement in Bayesian non-parametric mixture models', *Journal of the Royal Statistical Society Series B: Statistical Methodology* **69**(4), 715–740.

Luong, M., Nguyen, K., Phung, D., Haffari, G. & Qu, L. (2025), Unbiased sliced Wasserstein kernels for high-quality audio captioning, *in* 'The Thirty-ninth Annual Conference on Neural Information Processing Systems'.
**URL:** *https://openreview.net/forum?id=83HlJrZigZ*

Macchi, O. (1975), 'The coincidence approach to stochastic point processes', *Advances in Applied Probability* **7**(1), 83–122.

McAlinn, K. & Takanashi, K. (2026), 'When is generalized Bayes Bayesian? A decision-theoretic characterization of loss-based updating'.
**URL:** *https://arxiv.org/abs/2602.01573*

Meunier, D., Pontil, M. & Ciliberto, C. (2022), Distribution regression with sliced Wasserstein kernels, *in* 'International Conference on Machine Learning', PMLR, pp. 15501–15523.

Nadjahi, K., Durmus, A., Chizat, L., Kolouri, S., Shahrampour, S. & Simsekli, U. (2020), 'Statistical and topological properties of sliced probability divergences', *Advances in Neural Information Processing Systems* **33**, 20802–20812.

Neal, R. M. (2000), 'Markov chain sampling methods for Dirichlet process mixture models', *Journal of Computational and Graphical Statistics* **9**(2), 249–265.

Nguyen, K. (2025), 'An introduction to sliced optimal transport: Foundations, advances, extensions, and applications', *Foundations and Trends® in Computer Graphics and Vision* **17**(3-4), 171–391.

Nguyen, K., Bariletto, N. & Ho, N. (2024), Quasi-Monte Carlo for 3d sliced Wasserstein, *in* 'The Twelfth International Conference on Learning Representations'.

Nguyen, K. & Mueller, P. (2026), 'Summarizing nonparametric Bayesian mixture posteriors—sliced optimal transport metrics for Gaussian mixtures', *Journal of Computational and Graphical Statistics* pp. 1–12.

Okano, R. & Imaizumi, M. (2025), 'Wasserstein k-centers clustering for distributional data', *Statistics and Computing* **35**(5), 124.

Pedroso, R. C., Quintana, F. A. & Loschi, R. H. (2026), 'Bayesian repulsive mixture model for multivariate functional data', *arXiv preprint arXiv:2603.07320* .

Petralia, F., Rao, V. & Dunson, D. (2012), Repulsive mixtures, *in* F. Pereira, C. Burges, L. Bottou & K. Weinberger, eds, 'Advances in Neural Information Processing Systems', Vol. 25, Curran Associates, Inc.

Peyré, G. & Cuturi, M. (2019), 'Computational optimal transport: With applications to data science', *Foundations and Trends in Machine Learning* **11**(5-6), 355–607.

Peyré, G., Cuturi, M. et al. (2019), 'Computational optimal transport: With applications to data science', *Foundations and Trends in Machine Learning* **11**(5-6), 355–607.

Pitman, J. & Yor, M. (1997), 'The two-parameter Poisson-Dirichlet distribution derived from a stable subordinator', *The Annals of Probability* pp. 855–900.

Rabin, J., Ferradans, S. & Papadakis, N. (2014), Adaptive color transfer with relaxed optimal transport, *in* '2014 IEEE International Conference on Image Processing (ICIP)', IEEE, pp. 4852–4856.

Rabin, J., Peyré, G., Delon, J. & Bernot, M. (2012), Wasserstein barycenter and its application to texture mixing, *in* 'Scale Space and Variational Methods in Computer Vision: Third International Conference, SSVM 2011, Ein-Gedi, Israel, May 29–June 2, 2011, Revised Selected Papers 3', Springer, pp. 435–446.

Regazzini, E., Lijoi, A. & Prünster, I. (2003), 'Distributional results for means of normalized random measures with independent increments', *The Annals of Statistics* **31**(2), 560–585.

Rodriguez, A., Dunson, D. B. & Gelfand, A. E. (2008), 'The nested Dirichlet process', *Journal of the American Statistical Association* **103**(483), 1131–1154.

Rousseau, J. & Mengersen, K. (2011), 'Asymptotic behaviour of the posterior distribution in overfitted mixture models', *Journal of the Royal Statistical Society Series B: Statistical Methodology* **73**(5), 689–710.

Song, Z., Camerlenghi, F., Shen, W., Guindani, M. & Beraha, M. (2025), 'Repulsive mixture model with projection determinantal point process', *arXiv preprint arXiv:2510.08838* .

Steinwart, I. & Scovel, C. (2012), 'Mercer's theorem on general domains: On the interaction between measures, kernels, and RKHSs', *Constructive Approximation* **35**(3), 363–417.

Villani, C. (2009), *Optimal transport: old and new*, Vol. 338, Springer.

Xie, F. & Xu, Y. (2020), 'Bayesian repulsive Gaussian mixture model', *Journal of the American Statistical Association* **115**(529), 187–203.

Xu, Y., Müller, P. & Telesca, D. (2016), 'Bayesian inference for latent biologic structure with determinantal point processes (DPP)', *Biometrics* **72**(3), 955–964.

Yazar, S., Alquicira-Hernandez, J., Wing, K., Senabouth, A., Gordon, M. G., Andersen, S., Lu, Q., Rowson, A., Taylor, T. R., Clarke, L. et al. (2022), 'Single-cell eQTL mapping identifies cell type–specific genetic control of autoimmune disease', *Science* **376**(6589), eabf3041.

Zhuang, Y., Chen, X. & Yang, Y. (2022), 'Wasserstein $k$-means for clustering probability distributions', *Advances in Neural Information Processing Systems* **35**, 11382–11395.

# Supplementary for "Distributional Determinantal Point Process for Repulsive Clustering of Distributions"

## A Proofs

### A.1 Proof of Theorem 1

We define $T_L : \mathbb{L}_2(\Theta, P_0) \to \mathbb{L}_2(\Theta, P_0)$ be the integral operator of $L$:

$$(T_L f)(\nu) = \int_{\Theta} L(\nu, \nu') f(\nu') \mathrm{d}P_0(\nu'), \tag{12}$$

where $\mathbb{L}_2(\Theta, P_0) = \left\{ f : \Theta \to \mathbb{R} \mid \int_{\Theta} |f(\nu)|^2 \,\mathrm{d}P_0(\nu) < \infty \right\}$. We need to show two properties to prove that dDPP is a well-defined point process (i) $L(\nu, \nu')$ is positive definite, (ii) $T_L$ to be trace-class on $\mathbb{L}_2(\Theta, P_0)$ which leads to $\sum_{k=0}^{\infty} \frac{1}{k!} \int_{\Theta^k} \det \left( L(\nu_i, \nu_j) \right)_{i,j=1}^{k} \mathrm{d}P_0(\nu_1) \ldots \mathrm{d}P_0(\nu_k) = \det(I + T_L)$ to be finite.

**Symmetric positive definiteness.** The kernel is symmetric due to the symmetry of SW distance. We only need to prove that it is positive definite. Let $\Theta \subset \mathcal{P}_2(\mathbb{R}^d)$ and define $\Psi[\nu](\theta, t) = F^{-1}_{\theta\sharp\nu}(t)$, for $(\theta, t) \in \mathbb{S}^{d-1} \times [0,1]$. From the quantile representation of SW distance, we have:

$$\mathrm{SW}_2^2(\nu, \nu') = \|\Psi(\nu) - \Psi(\nu')\|^2_{\mathbb{L}_2(\mathbb{S}^{d-1} \times [0,1])}. \tag{13}$$

For any $\nu_1, \ldots, \nu_k \in \Theta$ and $c_1, \ldots, c_k \in \mathbb{R}$, we need to show

$$\sum_{i=1}^{k} \sum_{j=1}^{k} c_i c_j L(\nu_i, \nu_j) \geq 0. \tag{14}$$

Define $f_i = \Psi(\nu_i) \in \mathbb{L}_2(\mathbb{S}^{d-1} \times [0,1])$, we then have $L(\nu_i, \nu_j) = \exp\left( -\gamma \|f_i - f_j\|^2 \right)$. Using the identity

$$\|f_i - f_j\|^2 = \|f_i\|^2 + \|f_j\|^2 - 2\langle f_i, f_j \rangle, \tag{15}$$

we obtain

$$L(\nu_i, \nu_j) = \exp(-\gamma\|f_i\|^2)\exp(-\gamma\|f_j\|^2)\exp\Big(2\gamma\langle f_i, f_j\rangle\Big). \tag{16}$$

Thus,

$$\sum_{i,j=1}^{k} c_i c_j L(\nu_i, \nu_j) = \sum_{i,j=1}^{k}\Big(c_i e^{-\gamma\|f_i\|^2}\Big)\Big(c_j e^{-\gamma\|f_j\|^2}\Big)\exp\Big(2\gamma\langle f_i, f_j\rangle\Big). \tag{17}$$

Let $a_i = c_i e^{-\gamma\|f_i\|^2}$, it suffices to show

$$\sum_{i=1}^{k}\sum_{j=1}^{k} a_i a_j \exp\Big(2\gamma\langle f_i, f_j\rangle\Big) \geq 0. \tag{18}$$

We observe that the kernel

$$\exp\Big(2\gamma\langle f_i, f_j\rangle\Big) \tag{19}$$

is positive definite on a Hilbert space. Using the power series expansion, we have $\exp\Big(2\gamma\langle f_i, f_j\rangle\Big) = \sum_{m=0}^{\infty}\frac{(2\gamma)^m}{m!}\langle f_i, f_j\rangle^m$, and each kernel $\langle f_i, f_j\rangle^m$ is positive definite. Since nonnegative linear combinations of positive definite kernels are positive definite, we have

$$\sum_{i,j=1}^{k} a_i a_j \exp\Big(2\gamma\langle f_i, f_j\rangle\Big) \geq 0, \tag{20}$$

which implies

$$\sum_{i,j=1}^{k} c_i c_j L(\nu_i, \nu_j) \geq 0. \tag{21}$$

Hence $L$ is a positive definite kernel.

**(ii) Trace-class.** We prove that $T_L$ is trace-class via Mercer's theorem. To apply Mercer's theorem (Steinwart & Scovel 2012), we need $\Theta$ to be compact (given by Assumption 1) and $L$ to be continuous on $\Theta\times\Theta$. We establish continuity as follows. The map $(\nu,\nu') \to \mathrm{SW}_2^2(\nu,\nu')$ is continuous on $\Theta\times\Theta$, since for any sequences $\nu_n \to \nu$ and $\nu_n' \to \nu'$, we have

$$|\mathrm{SW}_2^2(\nu_n,\nu_n') - \mathrm{SW}_2^2(\nu,\nu')| \leq |\mathrm{SW}_2(\nu_n,\nu_n') - \mathrm{SW}_2(\nu,\nu')|\Big(\mathrm{SW}_2(\nu_n,\nu_n') + \mathrm{SW}_2(\nu,\nu')\Big) \tag{22}$$

$$\leq C\Big(\mathrm{SW}_2(\nu_n,\nu) + \mathrm{SW}_2(\nu_n',\nu')\Big) \to 0, \tag{23}$$

where the second inequality uses the triangle inequality for $\mathrm{SW}_2$, the convergence to zero follows from (Nadjahi et al. 2020), and $C < \infty$ is the finite diameter of $\Theta$ under $\mathrm{SW}_2$, which follows from compactness. Since $x \mapsto \exp(-\gamma x)$ is continuous, the kernel

$$L(\nu, \nu') = \exp(-\gamma \mathrm{SW}_2^2(\nu, \nu')) \tag{24}$$

is continuous on $\Theta \times \Theta$.

Therefore, since $P_0$ is a Borel probability measure on $\Theta$, Mercer's theorem applies and $T_L$ admits an orthonormal eigenexpansion with non-negative eigenvalues $\{\lambda_i\}_{i=1}^{\infty}$, and its trace satisfies

$$\mathrm{Tr}(T_L) = \int_\Theta L(\nu, \nu)\, \mathrm{d}P_0(\nu). \tag{25}$$

Since $\mathrm{SW}_2(\nu, \nu) = 0$ for all $\nu \in \Theta$, we have $L(\nu, \nu) = 1$, and hence

$$\mathrm{Tr}(T_L) = \int_\Theta 1\, \mathrm{d}P_0(\nu) = 1. \tag{26}$$

Since $L$ is positive definite (shown in part (i)), all eigenvalues satisfy $\lambda_i \geq 0$, and therefore

$$\sum_{i=1}^{\infty} \lambda_i = \mathrm{Tr}(T_L) = 1 < \infty, \tag{27}$$

which by definition establishes that $T_L$ is trace-class. Consequently, the Fredholm determinant admits the spectral representation

$$\det(I + T_L) = \prod_{i=1}^{\infty} (1 + \lambda_i). \tag{28}$$

Using the inequality $\log(1 + \lambda_i) \leq \lambda_i$ for $\lambda_i \geq 0$, we obtain

$$\log\det(I + T_L) = \sum_{i=1}^{\infty} \log(1 + \lambda_i) \leq \sum_{i=1}^{\infty} \lambda_i = \mathrm{Tr}(T_L) = 1 < \infty, \tag{29}$$

and hence $\det(I + T_L) \leq e < \infty$, which completes the proof.

## A.2 Proof of Theorem 2

Since the map $x \to e^{-\gamma x}$ is differentiable with derivative $-\gamma e^{-\gamma x}$, the mean value theorem yields

$$|\hat{L}_{ij} - L_{ij}| = \gamma\, e^{-\gamma \xi_{ij}} \left| \mathrm{SW}_2^2\left(\hat{\nu}_i^{(m)}, \hat{\nu}_j^{(m)}\right) - \mathrm{SW}_2^2(\nu_i, \nu_j) \right| \tag{30}$$

for some $\xi_{ij}$ lying between $\mathrm{SW}_2^2(\hat{\nu}_i^{(m)}, \hat{\nu}_j^{(m)})$ and $\mathrm{SW}_2^2(\nu_i, \nu_j)$. Since $e^{-\gamma \xi_{ij}} \leq 1$, we have

$$|\hat{L}_{ij} - L_{ij}| \leq \gamma \left| \mathrm{SW}_2^2\left(\hat{\nu}_i^{(m)}, \hat{\nu}_j^{(m)}\right) - \mathrm{SW}_2^2(\nu_i, \nu_j) \right|. \tag{31}$$

It therefore suffices to bound $|\mathrm{SW}_2^2(\hat{\nu}_i^{(m)}, \hat{\nu}_j^{(m)}) - \mathrm{SW}_2^2(\nu_i, \nu_j)|$ directly. Using the closed-form expression of $\mathrm{SW}_2^2$ and Jensen's inequality (twice):

$$\begin{aligned}
&\left| \mathrm{SW}_2^2\left(\hat{\nu}_i^{(m)}, \hat{\nu}_j^{(m)}\right) - \mathrm{SW}_2^2(\nu_i, \nu_j) \right| \\
&\quad = \left| \mathbb{E}_{v \sim \mathcal{U}(\mathbb{S}^{d-1})} \left[ \int_0^1 \left( \left( F^{-1}_{P_v \sharp \hat{\theta}_i^{o,(m)}}(t) - F^{-1}_{P_v \sharp \hat{\nu}_j^{(m)}}(t) \right)^2 - \left( F^{-1}_{P_v \sharp \nu_i}(t) - F^{-1}_{P_v \sharp \nu_j}(t) \right)^2 \right) \mathrm{d}t \right] \right| \\
&\quad \leq \mathbb{E}_{v \sim \mathcal{U}(\mathbb{S}^{d-1})} \left[ \int_0^1 \left| \left( F^{-1}_{P_v \sharp \hat{\theta}_i^{o,(m)}}(t) - F^{-1}_{P_v \sharp \hat{\nu}_j^{(m)}}(t) \right)^2 - \left( F^{-1}_{P_v \sharp \nu_i}(t) - F^{-1}_{P_v \sharp \nu_j}(t) \right)^2 \right| \mathrm{d}t \right].
\end{aligned} \tag{32}$$

For each fixed $v \in \mathbb{S}^{d-1}$ and $t \in [0,1]$, we define

$$p_v(t) := F^{-1}_{P_v \sharp \hat{\theta}_i^{o,(m)}}(t) - F^{-1}_{P_v \sharp \hat{\nu}_j^{(m)}}(t), \qquad q_v(t) := F^{-1}_{P_v \sharp \nu_i}(t) - F^{-1}_{P_v \sharp \nu_j}(t). \tag{33}$$

Both quantities are bounded in $[-2R, 2R]$ since $\mathrm{supp}(\nu_i) \subset B(0,R)$ implies $P_v \sharp \nu_i$ is supported on $[-R, R]$ for all $v$. Applying $|p_v(t)^2 - q_v(t)^2| = |p_v(t) - q_v(t)||p_v(t) + q_v(t)| \leq 4R|p_v(t) - q_v(t)|$ and the triangle inequality $|p_v(t) - q_v(t)| \leq |F^{-1}_{P_v \sharp \hat{\theta}_i^{o,(m)}}(t) - F^{-1}_{P_v \sharp \nu_i}(t)| + |F^{-1}_{P_v \sharp \hat{\nu}_j^{(m)}}(t) - F^{-1}_{P_v \sharp \nu_j}(t)|$, we have:

$$\begin{aligned}
&\int_0^1 |p_v(t)^2 - q_v(t)^2| \, \mathrm{d}t \\
&\quad \leq 4R \left( \int_0^1 \left| F^{-1}_{P_v \sharp \hat{\theta}_i^{o,(m)}}(t) - F^{-1}_{P_v \sharp \nu_i}(t) \right| \mathrm{d}t + \int_0^1 \left| F^{-1}_{P_v \sharp \hat{\nu}_j^{(m)}}(t) - F^{-1}_{P_v \sharp \nu_j}(t) \right| \mathrm{d}t \right) \\
&\quad = 4R \left( W_1\left( P_v \sharp \hat{\theta}_i^{o,(m)}, P_v \sharp \nu_i \right) + W_1\left( P_v \sharp \hat{\nu}_j^{(m)}, P_v \sharp \nu_j \right) \right),
\end{aligned} \tag{34}$$

where we used the quantile representation $W_1(\mu,\nu) = \int_0^1 |F_\mu^{-1}(t) - F_\nu^{-1}(t)|\,\mathrm{d}t$. For measures on $[-R,R]$, $W_1(\mu,\nu) = \int_{\mathbb{R}} |F_\mu(x) - F_\nu(x)|\,dx \le 2R\|F_\mu - F_\nu\|_\infty$. Therefore (34) gives:

$$\int_0^1 |p_v(t)^2 - q_v(t)^2|\,\mathrm{d}t \le 8R^2\left(\left\|F_{P_v\sharp\hat{\theta}_i^{o,(m)}} - F_{P_v\sharp\nu_i}\right\|_\infty + \left\|F_{P_v\sharp\hat{\nu}_j^{(m)}} - F_{P_v\sharp\nu_j}\right\|_\infty\right). \tag{35}$$

Substituting (35) into (32) and defining, for each atom $i \in [Q]$, the projection-averaged error is

$$D_i := \mathbb{E}_{v\sim\mathcal{U}(\mathbb{S}^{d-1})}\left[\left\|F_{P_v\sharp\hat{\theta}_i^{o,(m)}} - F_{P_v\sharp\nu_i}\right\|_\infty\right]. \tag{36}$$

we the have

$$\left|\mathrm{SW}_2^2\left(\hat{\nu}_i^{(m)}, \hat{\nu}_j^{(m)}\right) - \mathrm{SW}_2^2(\nu_i,\nu_j)\right| \le 8R^2(D_i + D_j). \tag{37}$$

Next, we bound each $D_i$ in high probability through a Dvoretzky–Kiefer–Wolfowitz (DKW) expectation bound applied pointwise in $v$, followed by a bounded-differences concentration inequality. For each fixed $v \in \mathbb{S}^{d-1}$, the projected samples $\{v^\top x_t^{(i)}\}_{t=1}^n$ are i.i.d. from $P_v\sharp\nu_i$, and $F_{P_v\sharp\hat{\theta}_i^{o,(m)}}$ is their empirical CDF. The DKW inequality $\mathbb{P}(\|F_{P_v\sharp\hat{\theta}_i^{o,(m)}} - F_{P_v\sharp\nu_i}\|_\infty > \epsilon) \le 2e^{-2m\epsilon^2}$ gives

$$\mathbb{E}\left[\left\|F_{P_v\sharp\hat{\theta}_i^{o,(m)}} - F_{P_v\sharp\nu_i}\right\|_\infty\right] = \int_0^\infty \mathbb{P}(\|\cdot\|_\infty > \epsilon)\,d\epsilon \le \int_0^\infty 2e^{-2m\epsilon^2}\,d\epsilon = \sqrt{\frac{\pi}{2m}}, \tag{38}$$

uniformly in $v$. By Fubini's theorem, we have

$$\mathbb{E}[D_i] = \mathbb{E}_v\,\mathbb{E}\left[\left\|F_{P_v\sharp\hat{\theta}_i^{o,(m)}} - F_{P_v\sharp\nu_i}\right\|_\infty\right] \le \sqrt{\frac{\pi}{2m}}. \tag{39}$$

Viewing $D_i$ as a function of the $n$ i.i.d. samples $\{x_t^{(i)}\}_{t=1}^m$, replacing a single sample changes each empirical CDF $F_{P_v\sharp\hat{\theta}_i^{o,(m)}}$ by at most $1/m$ in sup-norm, hence changes $\|F_{P_v\sharp\hat{\theta}_i^{o,(m)}} - F_{P_v\sharp\nu_i}\|_\infty$ and its $\mathbb{E}_v$-average $D_i$ by at most $1/m$. McDiarmid's inequality therefore gives $\mathbb{P}(D_i - \mathbb{E}[D_i] \ge s) \le \exp(-2ms^2)$, so with probability at least $1-\delta_0$,

$$D_i \le \mathbb{E}[D_i] + \sqrt{\frac{\log(1/\delta_0)}{2m}} \le \sqrt{\frac{\pi}{2m}} + \sqrt{\frac{\log(1/\delta_0)}{2m}}. \tag{40}$$

Combining (37), (40) (for both $i$ and $j$) and (31), with probability at least $1-2\delta_0$:

$$|\hat{L}_{ij} - L_{ij}| \leq 16\gamma R^2 \left( \sqrt{\frac{\pi}{2m}} + \sqrt{\frac{\log(1/\delta_0)}{2m}} \right). \tag{41}$$

There are $Q(Q-1) \leq Q^2$ off-diagonal pairs $(i,j)$ (the diagonal entries satisfy $\hat{L}_{ii} = L_{ii} = 1$). Setting $\delta_0 = \delta/(2Q^2)$ and applying a union bound over all pairs, with probability at least $1-\delta$, we have for all $(i,j) \in [Q]^2$:

$$|\hat{L}_{ij} - L_{ij}| \leq 16\gamma R^2 \left( \sqrt{\frac{\pi}{2m}} + \sqrt{\frac{\log(2Q^2/\delta)}{2m}} \right). \tag{42}$$

Taking the maximum and applying $\|\cdot\|_2 \leq \|\cdot\|_F \leq Q\|\cdot\|_{\max}$, together with $\sqrt{a}+\sqrt{b} \leq \sqrt{2}\sqrt{a+b}$, we have

$$\|\hat{\mathbf{L}} - \mathbf{L}\|_2 \leq \frac{16\gamma R^2 Q}{\sqrt{2m}} \left( \sqrt{\pi} + \sqrt{\log(2Q^2/\delta)} \right) \leq \frac{16\gamma R^2 Q}{\sqrt{m}} \sqrt{\pi + \log(2Q^2/\delta)}, \tag{43}$$

establishing the bound for $\mathbf{L}$.

**Propagation to $\hat{\mathbf{K}}$ via Resolvent Identity.** We propagate through the map $\mathbf{K} = \mathbf{L}(I_Q + \mathbf{L})^{-1}$. For any two matrices $A, B \succ -I$, the resolvent identity gives

$$A(I+A)^{-1} - B(I+B)^{-1} = (I+A)^{-1}(A-B)(I+B)^{-1}. \tag{44}$$

Applying this with $A = \hat{\mathbf{L}}$ and $B = \mathbf{L}$, we have

$$\hat{\mathbf{K}} - \mathbf{K} = (I+\hat{\mathbf{L}})^{-1}(\hat{\mathbf{L}} - \mathbf{L})(I+\mathbf{L})^{-1}. \tag{45}$$

Taking spectral norms and applying submultiplicativity, we have

$$\|\hat{\mathbf{K}} - \mathbf{K}\|_2 \leq \|(I+\hat{\mathbf{L}})^{-1}\|_2 \|\hat{\mathbf{L}} - \mathbf{L}\|_2 \|(I+\mathbf{L})^{-1}\|_2. \tag{46}$$

Since $\mathbf{L} \succ 0$, we have $\|(I+\mathbf{L})^{-1}\|_2 = (1+\lambda_{\min}(\mathbf{L}))^{-1} =: \kappa < 1$, and since $\hat{\mathbf{L}} \succ 0$, we have $\|(I+\hat{\mathbf{L}})^{-1}\|_2 \leq 1$. Substituting and applying (43):

$$\|\hat{\mathbf{K}} - \mathbf{K}\|_2 \leq \frac{16\kappa\gamma R^2 Q}{\sqrt{m}} \sqrt{\pi + \log(2Q^2/\delta)}. \tag{47}$$

This completes the proof of Theorem 2.

### A.3 Proof of Corollary 1

We denote $\mathbf{A}$ as either $\mathbf{L}_S$ or $\mathbf{K}_S$, and write $\hat{\mathbf{A}} = \mathbf{A} + \mathbf{E}$ where $\mathbf{E} = \hat{\mathbf{A}} - \mathbf{A}$. We denote the rows of $\mathbf{A}$ and $\mathbf{E}$ by $\mathbf{a}_1, \ldots, \mathbf{a}_k$ and $\mathbf{e}_1, \ldots, \mathbf{e}_k$ respectively.

Since the determinant is linear in each row separately, applying linearity to each of the $k$ rows of $\mathbf{A} + \mathbf{E}$ produces $2^k$ terms, one for each subset $T \subseteq [k]$ recording which rows are taken from $\mathbf{E}$:

$$\det(\mathbf{A} + \mathbf{E}) = \sum_{T \subseteq [k]} \det(\mathbf{M}_T), \tag{48}$$

where $\mathbf{M}_T$ is the $k \times k$ matrix with rows

$$(\mathbf{M}_T)_a = \begin{cases} \mathbf{e}_a & a \in T, \\ \mathbf{a}_a & a \notin T. \end{cases} \tag{49}$$

The term $T = \emptyset$ gives $\det(\mathbf{M}_\emptyset) = \det(\mathbf{A})$. Subtracting and grouping by $\ell = |T|$, we have

$$\det(\mathbf{A} + \mathbf{E}) - \det(\mathbf{A}) = \sum_{\ell=1}^{k} \sum_{\substack{T \subseteq [k] \\ |T| = \ell}} \det(\mathbf{M}_T). \tag{50}$$

Each $\mathbf{M}_T$ with $|T| = \ell$ has exactly $\ell$ rows from $\mathbf{E}$ and $k - \ell$ rows from $\mathbf{A}$. We apply Hadamard's inequality $|\det(\mathbf{M})| \leq \prod_{a=1}^{k} \|\mathbf{m}_a\|_2$ to each term:

$$|\det(\mathbf{M}_T)| \leq \prod_{a \in T} \|\mathbf{e}_a\|_2 \prod_{a \notin T} \|\mathbf{a}_a\|_2 \leq \|\mathbf{E}\|_2^{\ell} \|\mathbf{A}\|_2^{k-\ell}. \tag{51}$$

Since there are $\binom{k}{\ell}$ subsets $T$ of size $\ell$, taking absolute values and summing over all $\ell$:

$$\left|\det(\hat{\mathbf{A}}) - \det(\mathbf{A})\right| \leq \sum_{\ell=1}^{k} \binom{k}{\ell} \|\mathbf{E}\|_2^{\ell} \|\mathbf{A}\|_2^{k-\ell} = (\|\mathbf{A}\|_2 + \|\mathbf{E}\|_2)^k - \|\mathbf{A}\|_2^k. \tag{52}$$

Factoring out $\|\mathbf{A}\|_2^k$ and applying $(1+x)^k - 1 \leq kx\, e^{(k-1)x}$ for $x \geq 0$ with $x = \|\mathbf{E}\|_2 / \|\mathbf{A}\|_2$, we have:

$$\left|\det(\hat{\mathbf{A}}) - \det(\mathbf{A})\right| \leq k\, e^{(k-1)\|\mathbf{E}\|_2/\|\mathbf{A}\|_2} \|\mathbf{A}\|_2^{k-1} \|\mathbf{E}\|_2. \tag{53}$$

**Bound for** $\det(\hat{\mathbf{L}}_S)$**.** Applying the inequality $b^k - a^k \leq k(b-a)b^{k-1}$ for $0 \leq a \leq b$ to (52) with $a = \|\mathbf{L}_S\|_2$ and $b = \|\mathbf{L}_S\|_2 + \|\mathbf{E}\|_2$, we have:

$$\left|\det(\hat{\mathbf{L}}_S) - \det(\mathbf{L}_S)\right| \leq k\|\mathbf{E}\|_2 \left(\|\mathbf{L}_S\|_2 + \|\mathbf{E}\|_2\right)^{k-1}. \tag{54}$$

Since all entries of $\mathbf{L}_S$ and $\hat{\mathbf{L}}_S$ satisfy $L_{ij}, \hat{L}_{ij} \in (0,1]$ with unit diagonal, we have $\|\mathbf{L}_S\|_2 \leq \|\mathbf{L}_S\|_F \leq k$ and$\|\hat{\mathbf{L}}_S\|_2 \leq k$. Consequently, we have $\|\mathbf{E}\|_2 = \|\hat{\mathbf{L}}_S - \mathbf{L}_S\|_2 \leq \|\hat{\mathbf{L}}_S\|_2 + \|\mathbf{L}_S\|_2 \leq 2k$ and therefore $\|\mathbf{L}_S\|_2 + \|\mathbf{E}\|_2 \leq 3k$. Together with $\|\mathbf{E}\|_2 = \|\hat{\mathbf{L}}_S - \mathbf{L}_S\|_2 \leq \|\hat{\mathbf{L}} - \mathbf{L}\|_2$, we have:

$$\left|\det(\hat{\mathbf{L}}_S) - \det(\mathbf{L}_S)\right| \leq k(3k)^{k-1}\|\hat{\mathbf{L}} - \mathbf{L}\|_2. \tag{55}$$

Substituting the bound on $\|\hat{\mathbf{L}} - \mathbf{L}\|_2$ from Theorem 2, with probability at least $1-\delta$:

$$\left|\det(\hat{\mathbf{L}}_S) - \det(\mathbf{L}_S)\right| \leq \frac{16\,k(3k)^{k-1}\gamma R^2 Q}{\sqrt{m}}\sqrt{\pi + \log(2Q^2/\delta)}, \tag{56}$$

establishing (4).

**Bound for** $\det(\hat{\mathbf{K}}_S)$**.** Since all eigenvalues of $\mathbf{K}$ are of the form $\lambda_\ell/(1+\lambda_\ell) \in (0,1)$, we have $\|\mathbf{K}_S\|_2 \leq 1$ and similarly $\|\hat{\mathbf{K}}_S\|_2 \leq 1$ (as $\hat{\mathbf{K}} = \hat{\mathbf{L}}(I + \hat{\mathbf{L}})^{-1}$ has the same eigenvalue form). Applying $b^k - a^k \leq k(b-a)b^{k-1}$ to (52) with $a = \|\mathbf{K}_S\|_2$ and $b = \|\mathbf{K}_S\|_2 + \|\mathbf{E}\|_2$, and using $\|\mathbf{E}\|_2 = \|\hat{\mathbf{K}}_S - \mathbf{K}_S\|_2 \leq \|\hat{\mathbf{K}}_S\|_2 + \|\mathbf{K}_S\|_2 \leq 2$ so that $\|\mathbf{K}_S\|_2 + \|\mathbf{E}\|_2 \leq 3$:

$$\left|\det(\hat{\mathbf{K}}_S) - \det(\mathbf{K}_S)\right| \leq k\|\mathbf{E}\|_2 \left(\|\mathbf{K}_S\|_2 + \|\mathbf{E}\|_2\right)^{k-1} \leq k\,3^{k-1}\|\hat{\mathbf{K}} - \mathbf{K}\|_2, \tag{57}$$

where we used $\|\mathbf{E}\|_2 = \|\hat{\mathbf{K}}_S - \mathbf{K}_S\|_2 \leq \|\hat{\mathbf{K}} - \mathbf{K}\|_2$. Substituting the bound on $\|\hat{\mathbf{K}} - \mathbf{K}\|_2$ from Theorem 2, with probability at least $1-\delta$:

$$\left|\det(\hat{\mathbf{K}}_S) - \det(\mathbf{K}_S)\right| \leq \frac{16\,k\,3^{k-1}\,\kappa\gamma R^2 Q}{\sqrt{m}}\sqrt{\pi + \log(2Q^2/\delta)}, \tag{58}$$

establishing (5).

## A.4 Proof of Theorem 3

We first define the individual log generalized likelihood as follows:

$$r(F,\pi) := -\log\sum_{q=1}^{Q}\pi_q \exp\left(-w\,\mathrm{SW}_2^2(F,\nu_q)\right). \tag{59}$$

We then define the population risk $R(\pi) = \mathbb{E}_{F \sim G_0}[r(F,\pi)]$ and the empirical risk $\hat{R}_N(\pi) = \frac{1}{N}\sum_{i=1}^N r(F_i,\pi)$.

**Bounding the population risk and its continuity.** We first note that assumption (8) implies

$$\mathbb{E}_{F \sim G_0}\left[\mathrm{SW}_2^2(F,\nu_q)\right] < \infty \quad \text{for each } q, \tag{60}$$

since $w\,\mathrm{SW}_2^2(F,\nu_q) \le \exp(w\sum_{q'}\mathrm{SW}_2^2(F,\nu_{q'}))$, and the right-hand side is integrable by assumption.

Since $\sum_q \pi_q \exp(-w\,\mathrm{SW}_2^2(F,\nu_q)) \le \sum_q \pi_q = 1$, we have $r(F,\pi) \ge 0$. For the upper bound, since $\sum_q \pi_q \exp(-w\,\mathrm{SW}_2^2(F,\nu_q)) \ge \exp(-w\max_q \mathrm{SW}_2^2(F,\nu_q))$, we have

$$r(F,\pi) \le w \max_q \mathrm{SW}_2^2(F,\nu_q) \le w\sum_{q=1}^Q \mathrm{SW}_2^2(F,\nu_q). \tag{61}$$

Taking expectations and using (60), we have

$$R(\pi) \le w\sum_{q=1}^Q \mathbb{E}_{F\sim G_0}\left[\mathrm{SW}_2^2(F,\nu_q)\right] < \infty \quad \text{for all } \pi \in \Delta^{Q-1}. \tag{62}$$

The continuity of $R$ on $\Delta^{Q-1}$ follows from continuity of $\pi \mapsto r(F,\pi)$ for each fixed $F$, together with the dominated convergence theorem using the integrable dominating function $H(F) = w\sum_q \mathrm{SW}_2^2(F,\nu_q)$ (integrable by (60)).

Since $R$ is continuous on the compact set $\Delta^{Q-1}$, the set $\mathcal{M}^* := \arg\min_{\pi\in\Delta^{Q-1}} R(\pi)$ is non-empty and compact. For every $\varepsilon > 0$, the set $\{\pi : \tilde{d}(\pi,\mathcal{M}^*) \ge \varepsilon\}$ is compact and disjoint from $\mathcal{M}^*$, we define

$$\delta(\varepsilon) := \inf_{\pi : d(\pi,\mathcal{M}^*) > \varepsilon}\Big[R(\pi) - R^*\Big] > 0, \qquad R^* := \min_{\pi\in\Delta^{Q-1}} R(\pi). \tag{63}$$

**Uniform strong law of large numbers.** Given a fixed $\pi \in \Delta^{Q-1}$, the random variables $r(F_1,\pi), r(F_2,\pi), \ldots$ are i.i.d. with finite mean $R(\pi)$. By the strong law of large numbers, we have

$$\hat{R}_N(\pi) \overset{a.s.}{\to} R(\pi). \tag{64}$$

We extend (64) to uniform convergence via a finite-net argument. For any $\pi, \pi' \in \Delta^{Q-1}$ and any $F$, setting $f_q(F) := \exp(-w\,\mathrm{SW}_2^2(F, \nu_q)) \in (0,1]$, let $A := \sum_q \pi_q f_q(F)$ and $B := \sum_q \pi'_q f_q(F)$, both in $(0,1]$. Using the inequality $|\log(A/B)| \leq |A-B|/\min(A,B)$ (valid for $A, B > 0$):

$$|r(F,\pi) - r(F,\pi')| = \left|\log \frac{B}{A}\right| \leq \frac{|A-B|}{\min(A,B)} = \frac{\left|\sum_q(\pi_q - \pi'_q) f_q(F)\right|}{\min(A,\,B)}. \tag{65}$$

Since $f_q(F) \leq 1$, the numerator satisfies $|\sum_q(\pi_q - \pi'_q) f_q(F)| \leq \|\pi - \pi'\|_1$. For the denominator, $\min(A,B) \geq \min_q f_q(F) = \exp(-w \max_q \mathrm{SW}_2^2(F,\nu_q))$. Therefore:

$$\begin{aligned} |r(F,\pi) - r(F,\pi')| &\leq \|\pi - \pi'\|_1 \exp\left(w \max_q \mathrm{SW}_2^2(F,\nu_q)\right) \\ &\leq \|\pi - \pi'\|_1 \exp\left(w \sum_q \mathrm{SW}_2^2(F,\nu_q)\right). \end{aligned} \tag{66}$$

We define $C_1 := \mathbb{E}_{G_0}[\exp(w \sum_q \mathrm{SW}_2^2(F,\nu_q))] < \infty$ by assumption (8). Taking expectations in (66), we have

$$|R(\pi) - R(\pi')| \leq C_1 \|\pi - \pi'\|_1. \tag{67}$$

Since $\Delta^{Q-1}$ is compact, for any $\eta > 0$ there exists a finite $\eta$-net $\mathcal{N}_\eta = \{\pi^{(1)}, \ldots, \pi^{(M_\eta)}\} \subset \Delta^{Q-1}$ such that for every $\pi \in \Delta^{Q-1}$ there exists some $\pi^{(j)} \in \mathcal{N}_\eta$ with $\|\pi - \pi^{(j)}\|_1 \leq \eta$. For any $\pi$, pick its nearest net point $\pi^{(j)}$, by the triangle inequality, we have

$$|\hat{R}_N(\pi) - R(\pi)| \leq \underbrace{|\hat{R}_N(\pi) - \hat{R}_N(\pi^{(j)})|}_{\text{Term 1}} + \underbrace{|\hat{R}_N(\pi^{(j)}) - R(\pi^{(j)})|}_{\text{Term 2}} + \underbrace{|R(\pi^{(j)}) - R(\pi)|}_{\text{Term 3}}. \tag{68}$$

*Term 3.* By (67): $|R(\pi^{(j)}) - R(\pi)| \leq C_1 \eta$.

*Term 1.* By (66), averaging over $i$:

$$|\hat{R}_N(\pi) - \hat{R}_N(\pi^{(j)})| \leq \eta \frac{1}{N} \sum_{i=1}^{N} \exp\left(w \sum_q \mathrm{SW}_2^2(F_i,\nu_q)\right) =: \eta Z_N. \tag{69}$$

Since $\exp(w\sum_q \mathrm{SW}_2^2(F_i,\nu_q))$ are i.i.d. with finite mean $C_1<\infty$, the strong law of large numbers gives $Z_N \xrightarrow{a.s.} C_1$. Hence there exists an a.s.-finite random index $N_1$ such that for all $N \geq N_1$, $Z_N \leq 2C_1$ almost surely, giving $|\hat{R}_N(\pi) - \hat{R}_N(\pi^{(j)})| \leq 2C_1\,\eta$.

*Term 2.* By the pointwise strong law of large numbers (64) applied at each of the $M_\eta < \infty$ net points, we have

$$\max_{j=1,\dots,M_\eta} |\hat{R}_N(\pi^{(j)}) - R(\pi^{(j)})| \xrightarrow{a.s.} 0. \tag{70}$$

Taking the supremum over $\pi \in \Delta^{Q-1}$ in (68) and applying the bounds above, for all $N \geq N_1$ almost surely:

$$\sup_{\pi\in\Delta^{Q-1}} |\hat{R}_N(\pi) - R(\pi)| \leq 3C_1\,\eta + \max_{j=1,\dots,M_\eta} |\hat{R}_N(\pi^{(j)}) - R(\pi^{(j)})|. \tag{71}$$

Taking $\limsup_{N\to\infty}$ and using Term 2, we have

$$\limsup_{N\to\infty} \sup_{\pi\in\Delta^{Q-1}} |\hat{R}_N(\pi) - R(\pi)| \leq 3C_1\,\eta \quad \text{a.s.} \tag{72}$$

Since $\eta > 0$ was arbitrary, letting $\eta \to 0$ along a countable sequence gives

$$\sup_{\pi\in\Delta^{Q-1}} |\hat{R}_N(\pi) - R(\pi)| \overset{a.s.}{\to} 0. \tag{73}$$

Setting $\eta_\varepsilon := \delta(\varepsilon)/4 > 0$, there exists an a.s.-finite random index $N_0$ such that for all $N \geq N_0$:

$$\sup_{\pi\in\Delta^{Q-1}} |\hat{R}_N(\pi) - R(\pi)| \leq \eta_\varepsilon. \tag{74}$$

**Full support of the dDPP prior.** The dDPP draws a random configuration For any subconfiguration $S \subseteq \Theta$, the Janossy density satisfies

$$j_{|S|}(S) \propto \det\Big(L(\nu_p,\nu_q)\Big)_{\nu_p,\nu_q\in S} > 0, \tag{75}$$

since the SW kernel matrix is strictly positive definite on distinct points. Hence the dDPP places strictly positive probability on every subconfiguration of $\Theta$, including the full configuration $\Theta$.

Conditional on selecting all of $\Theta$, the weights $\pi_q = s_q / \sum_{q'} s_{q'}$ follow a Dirichlet$(a,\ldots,a)$ distribution, which has a positive Lebesgue density on the relative interior $\mathrm{ri}(\Delta^{Q-1})$ but assigns zero mass to the boundary $\partial\Delta^{Q-1}$. To handle the case in which $\mathcal{M}^*$ intersects $\partial\Delta^{Q-1}$ (i.e., the population risk minimizer may be degenerate), we argue as follows. For any open set $U \subseteq \Delta^{Q-1}$ with $U \neq \emptyset$, the set $U \cap \mathrm{ri}(\Delta^{Q-1})$ is non-empty (since $\mathrm{ri}(\Delta^{Q-1})$ is dense in $\Delta^{Q-1}$), and the Dirichlet $(a,\ldots,a)$ measure assigns positive mass to it. Therefore

$$\Pi(U) > 0 \quad \text{for every non-empty open } U \subseteq \Delta^{Q-1}. \tag{76}$$

In particular, $\Pi(U^*_\rho) > 0$ for every $\rho > 0$, where $U^*_\rho = \{\pi : \tilde{d}(\pi, \mathcal{M}^*) < \rho\}$.

**Connection between the posterior and the empirical risk.** The model specifies a likelihood for observation $F$ given $\pi$ as $p(F \mid \pi) \propto \sum_{q=1}^{Q} \pi_q \exp(-w\,\mathrm{SW}_2^2(F, \nu_q))$, so that $-\log p(F \mid \pi) = r(F, \pi)$ up to an additive constant independent of $\pi$. With prior $\Pi$, the posterior satisfies

$$\Pi_N(A \mid F_1, \ldots, F_N) = \frac{\int_A \exp\Big(-N\hat{R}_N(\pi)\Big)\,\mathrm{d}\Pi(\pi)}{\int_{\Delta^{Q-1}} \exp\Big(-N\hat{R}_N(\pi)\Big)\,\mathrm{d}\Pi(\pi)}, \tag{77}$$

for any measurable $A \subseteq \Delta^{Q-1}$.

**Bounding the posterior probability.** We fix $\varepsilon > 0$ and let $A_\varepsilon = \{\pi : \tilde{d}(\pi, \mathcal{M}^*) > \varepsilon\}$. We work on the a.s.-event $\{N \geq N_0\}$ where (74) holds.

For any $\pi \in A_\varepsilon$, applying (74) and (63), we have

$$\hat{R}_N(\pi) \geq R(\pi) - \eta_\varepsilon \geq R^* + \delta(\varepsilon) - \eta_\varepsilon = R^* + 3\eta_\varepsilon, \tag{78}$$

where the last equality uses $\delta(\varepsilon) = 4\eta_\varepsilon$. Therefore:

$$\int_{A_\varepsilon} \exp\Big(-N\hat{R}_N(\pi)\Big)\,\mathrm{d}\Pi(\pi) \leq \exp\left(-N(R^* + 3\eta_\varepsilon)\right). \tag{79}$$

We choose $\rho = \rho(\varepsilon) \in (0, \varepsilon)$ small enough so that $U^*_\rho = \{\pi : \tilde{d}(\pi, \mathcal{M}^*) < \rho\} \subseteq A^c_\varepsilon$ (which holds for any $\rho < \varepsilon$ since $A_\varepsilon^c = \{d(\pi, \mathcal{M}^*) \leq \varepsilon\})$ and, by continuity of $R$, we have

$$\sup_{\pi \in U_\rho^*} R(\pi) \leq R^* + \eta_\varepsilon. \tag{80}$$

For any $\pi \in U_\rho^*$, applying (74) and (80) leads to

$$\hat{R}_N(\pi) \leq R(\pi) + \eta_\varepsilon \leq R^* + 2\eta_\varepsilon. \tag{81}$$

Therefore, we have

$$\begin{aligned} \int_{\Delta^{Q-1}} \exp\left(-N\hat{R}_N(\pi)\right) \mathrm{d}\Pi(\pi) &\geq \int_{U_\rho^*} \exp\left(-N\hat{R}_N(\pi)\right) \mathrm{d}\Pi(\pi) \\ &\geq \Pi(U_\rho^*) \exp\left(-N(R^* + 2\eta_\varepsilon)\right). \end{aligned} \tag{82}$$

Dividing (79) by (82) and using (77), with $C := 1/\Pi(U_\rho^*) \in (0, \infty)$ (finite by full support of the prior) and $c(\varepsilon) := \eta_\varepsilon = \delta(\varepsilon)/4 > 0$ (positive by (63)), we final have

$$\Pi_N\left(\tilde{d}(\pi, \mathcal{M}^*) > \varepsilon \,\middle|\, F_1, \ldots, F_N\right) \leq C \exp\left(-c(\varepsilon)\, N\right) \overset{a.s.}{\to} 0 \quad \text{as } N \to \infty, \tag{83}$$

which completes the proof.

# B Additional Experiments

In this section we provide additional qualitative results that complement the main text. All figures below use the point estimates of the mixing measure obtained from the decision-theoretic summarization of Section 5.3, and the DP and dDPP models are run under identical settings, same atom set $\Theta = \mathcal{S}$ and the same generalized likelihood scale $w$, so that any difference is attributable solely to the prior.

## B.1 Single-Cell Data

Figure 6 repeats the UMAP comparison of Figure 1 for the two larger likelihood scales $w = 7 \times 10^5$ (first row) and $w = 10^6$ (second row). The qualitative contrast observed at

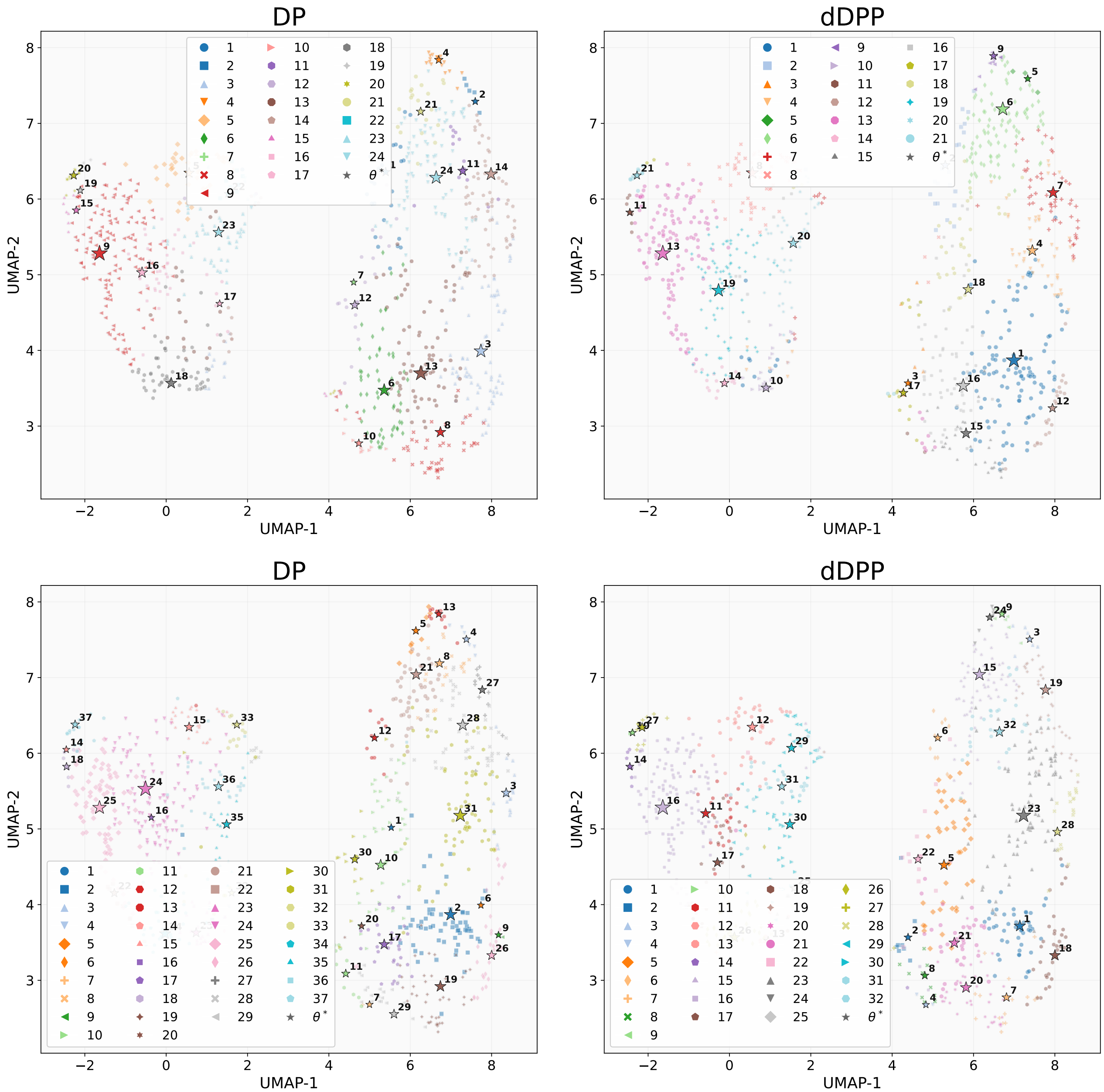



Figure 6: Same as Figure 1 with $7 \times 10^5$ (first row) and $w = 10^6$ (second row).

$w = 5 \times 10^5$ persists, and in fact sharpens, as $w$ increases: at every scale the dDPP atoms (right column) spread across distinct regions of the embedding and induce fewer, more spatially coherent groups, whereas the DP atoms (left column) concentrate more tightly and yield a more fragmented partition. This is consistent with Table 1, where the gap in cluster count between the two models widens with $w$: as the likelihood drives toward finer partitions, the repulsion in the dDPP prior increasingly suppresses redundant atoms.

Figures 7–10 display, for the dDPP solution at $w = 5 \times 10^5$, the donor-level cell distributions in the 17-dimensional PCA space for the individual clusters. Within each cluster, the donors exhibit visibly similar distributional shapes, and they closely resemble the corresponding representative atom shown in Figure 2. This indicates that the recovered atoms act as faithful prototypes for the donors assigned to them, and that the partition groups together donors with genuinely comparable gene-expression profiles rather than merely co-locating them in the embedding.

Figure 11 is the DP analogue of Figure 2, overlaying the recovered atoms on a KDE of the point-estimate mixing measure in the donor-level PCA space. Relative to the dDPP atoms, the DP atoms are placed closer together and several fall in overlapping high-density regions, so that distinct modes of the donor distribution are not as cleanly separated. This visual difference mirrors the higher (less negative) repulsion score attained by dDPP in Table 1.

## B.2 Human Epilepsy Project Data

Figure 12 repeats the UMAP comparison of Figure 3 for $w = 100$ (first row) and $w = 500$ (second row). As on the single-cell data, the dDPP solution (right column) assigns participants to fewer, more spatially coherent groups whose atoms are distributed across distinct regions of the embedding, while the DP solution (left column) clusters its atoms more tightly and produces a more fragmented partition. The effect is present at both the smaller and larger likelihood scales, confirming that the behavior reported at $w = 200$ in the main text is not an artifact of a particular choice of $w$.

Figures 13–15 visualize the recovered atoms as representative seizure-window patterns for both models at $w = 100$, $w = 200$, and $w = 500$ (DP in the first row, dDPP in the second row of each figure). Blue, red, and grey encode the daily states 0, 1, and 0.5 respectively, with yellow indicating padding for participants with shorter follow-up. Across all three scales, the dDPP atoms capture more clearly differentiated temporal profiles, ranging from seizure-free or near seizure-free patterns to denser, more irregular patterns, whereas the DP atoms include more nearly duplicated profiles, again reflecting the weaker separation quantified by the repulsion column of Table 2.

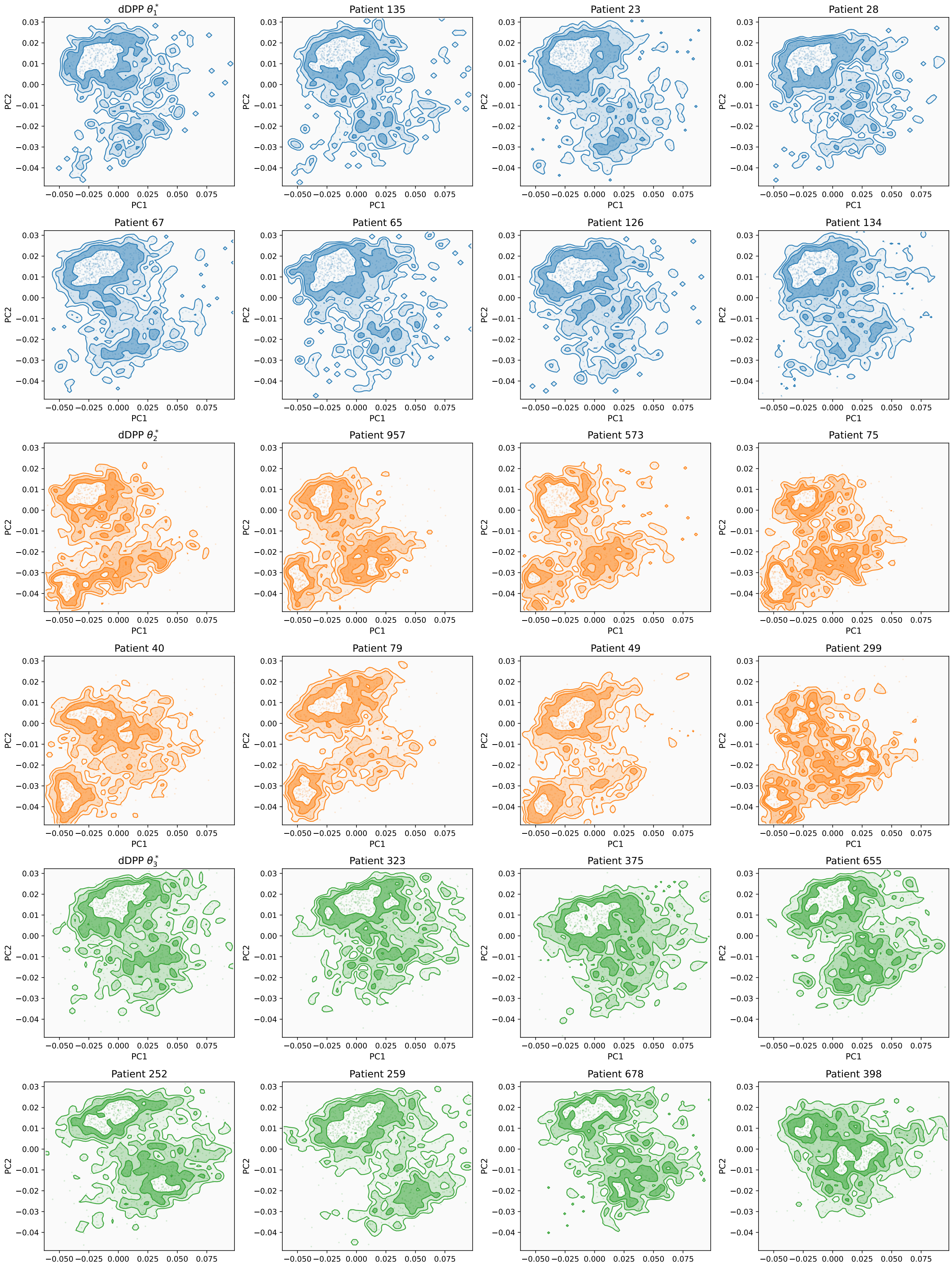



Figure 7: PCA visualization of some donors in clusters the dDPP model ($w = 5 \times 10^5, \gamma = 10^3$).

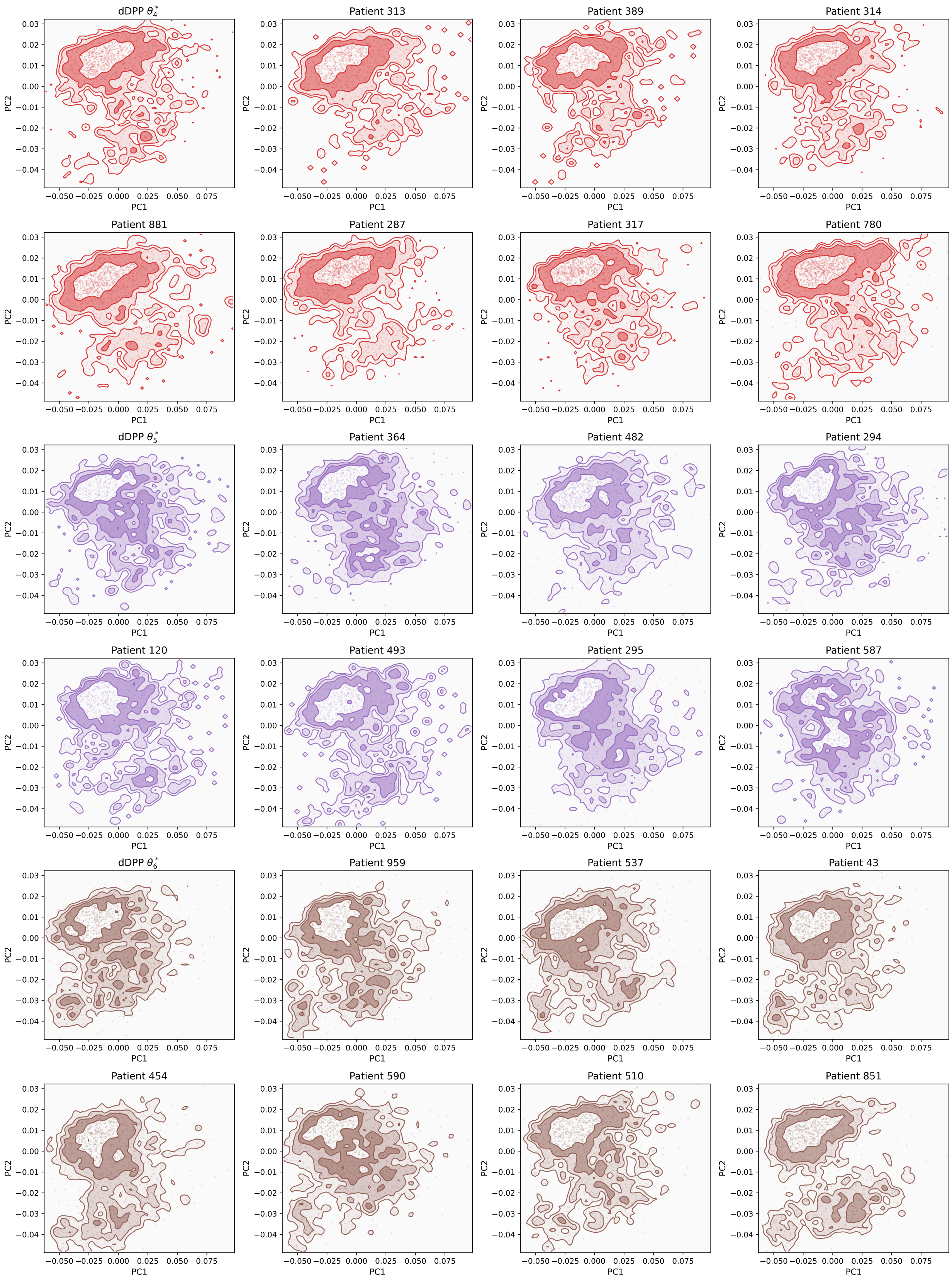



Figure 8: PCA visualization of some donors in clusters the dDPP model ($w = 5 \times 10^5, \gamma = 10^3$)

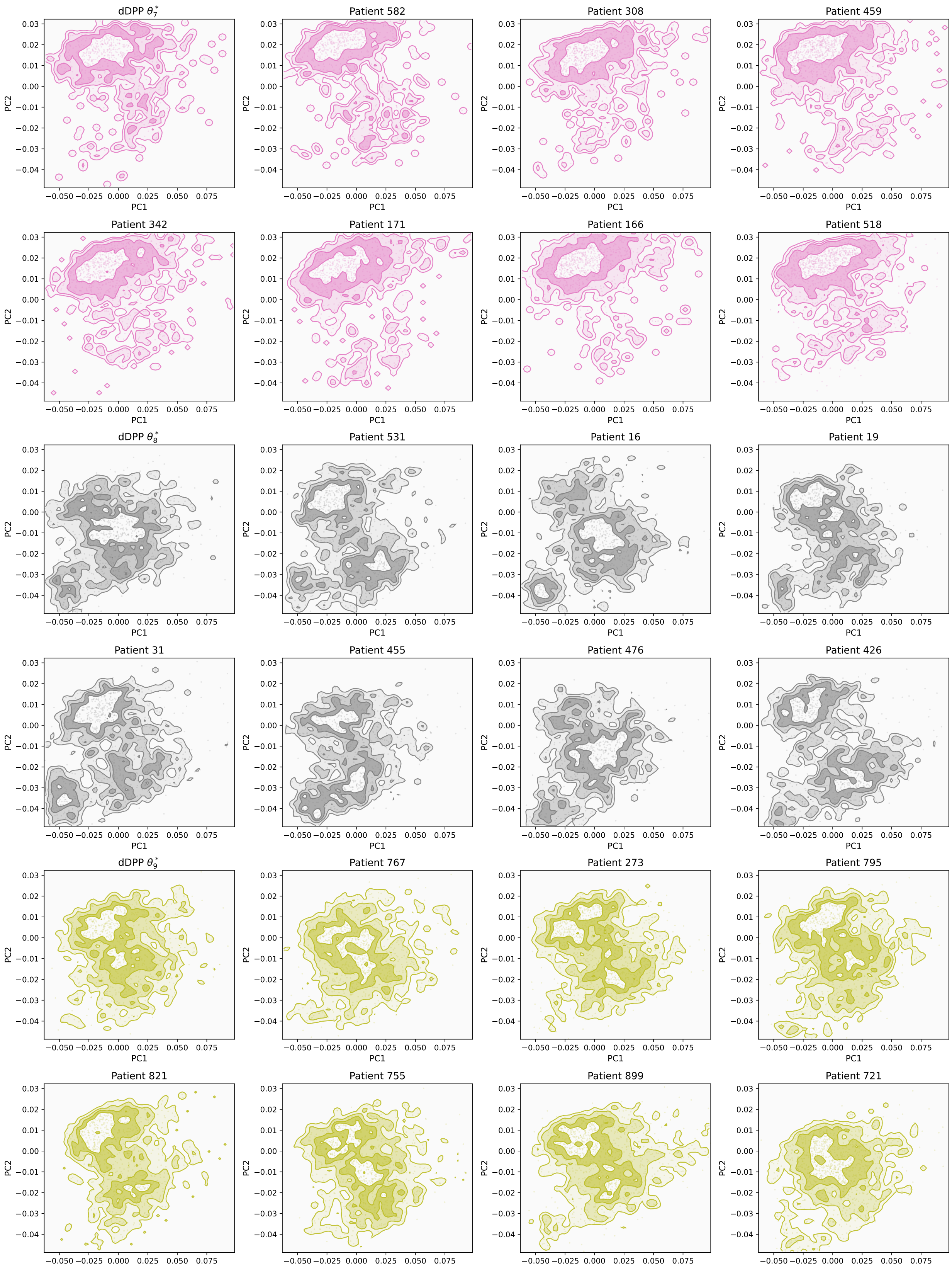



Figure 9: PCA visualization of some donors in clusters the dDPP model ($w = 5 \times 10^5, \gamma = 10^3$)

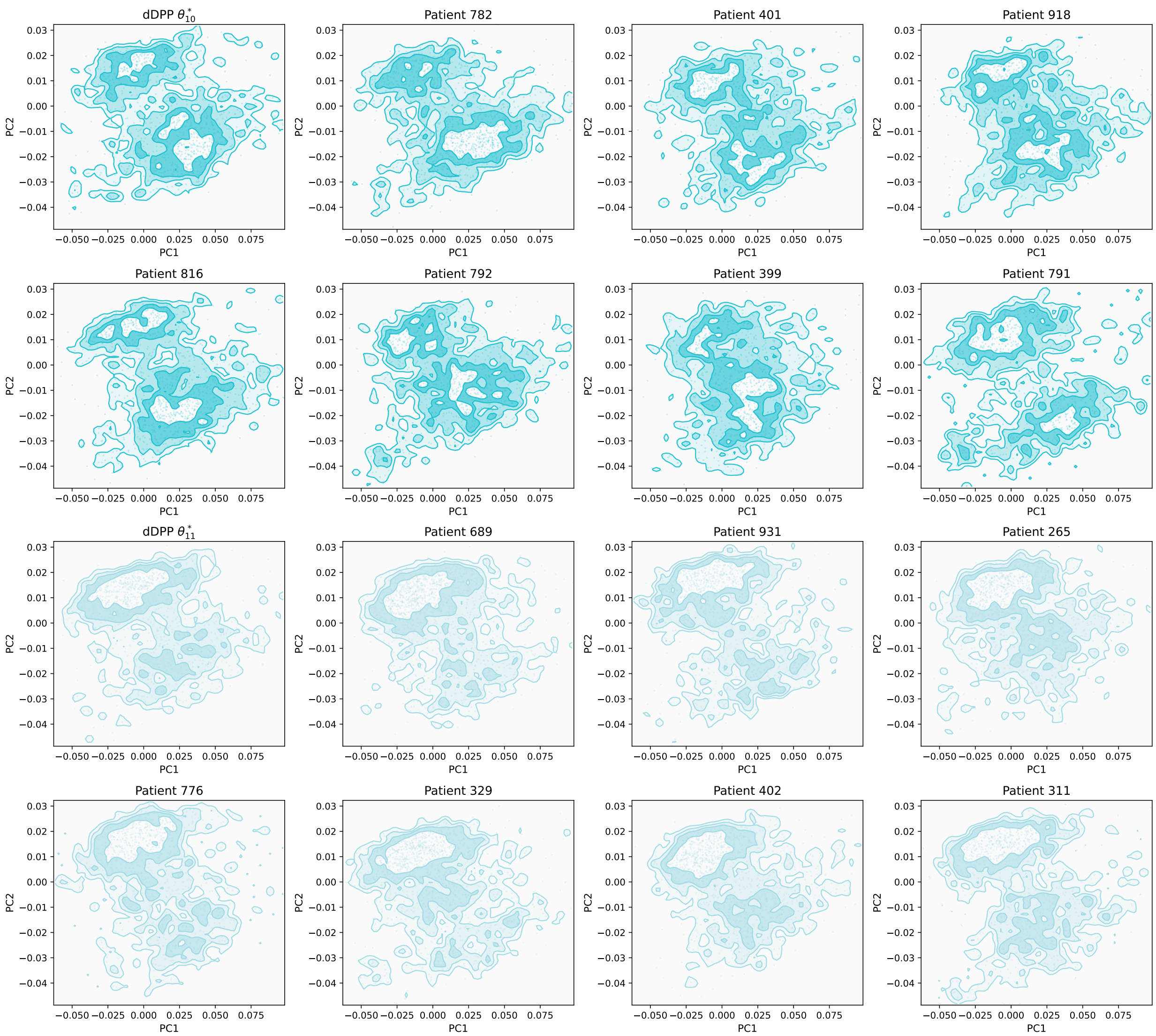



Figure 10: PCA visualization of some donors in clusters the dDPP model ($w = 5 \times 10^5, \gamma = 10^3$)

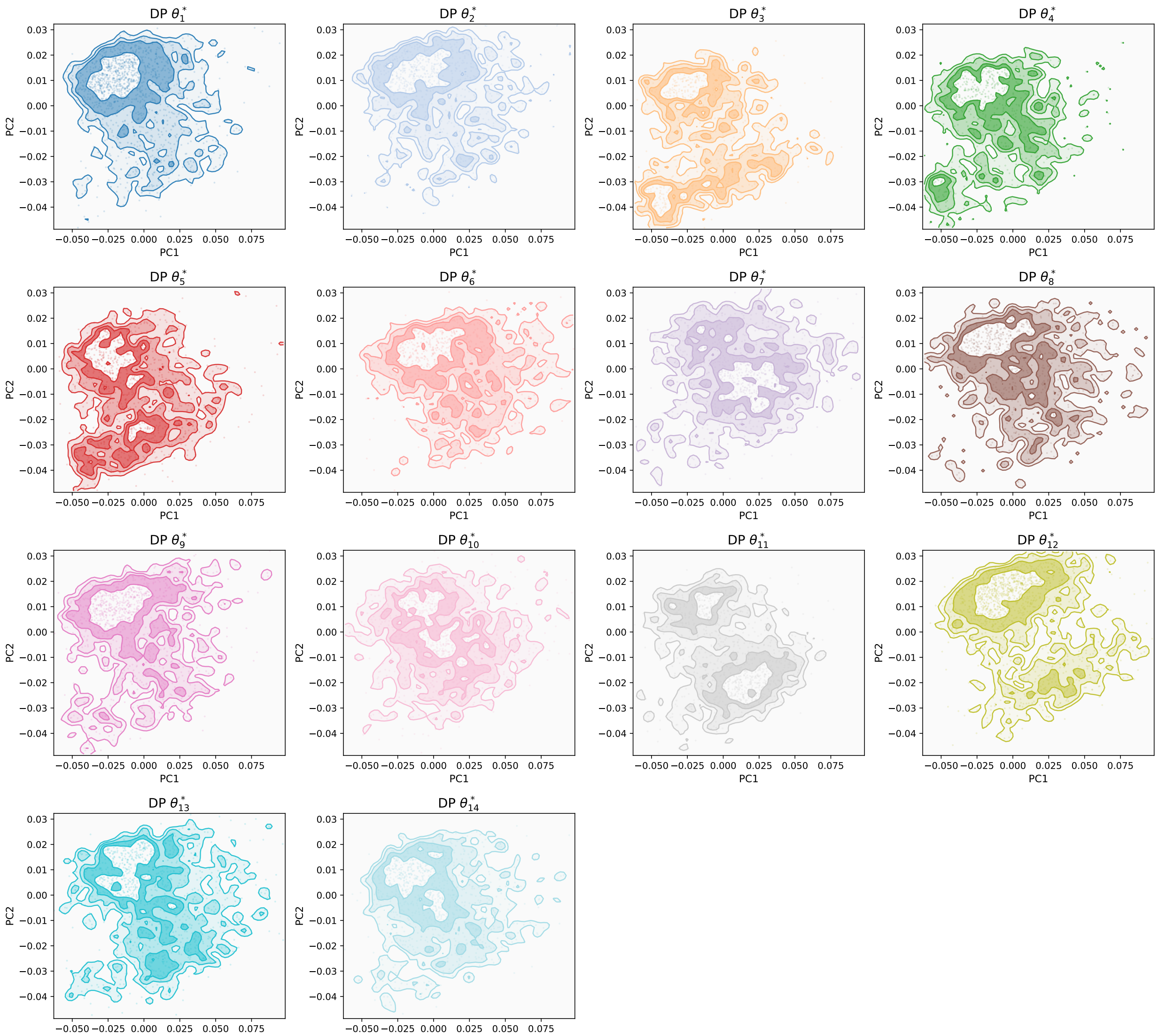



Figure 11: Same as Figure 2 for the DP model.

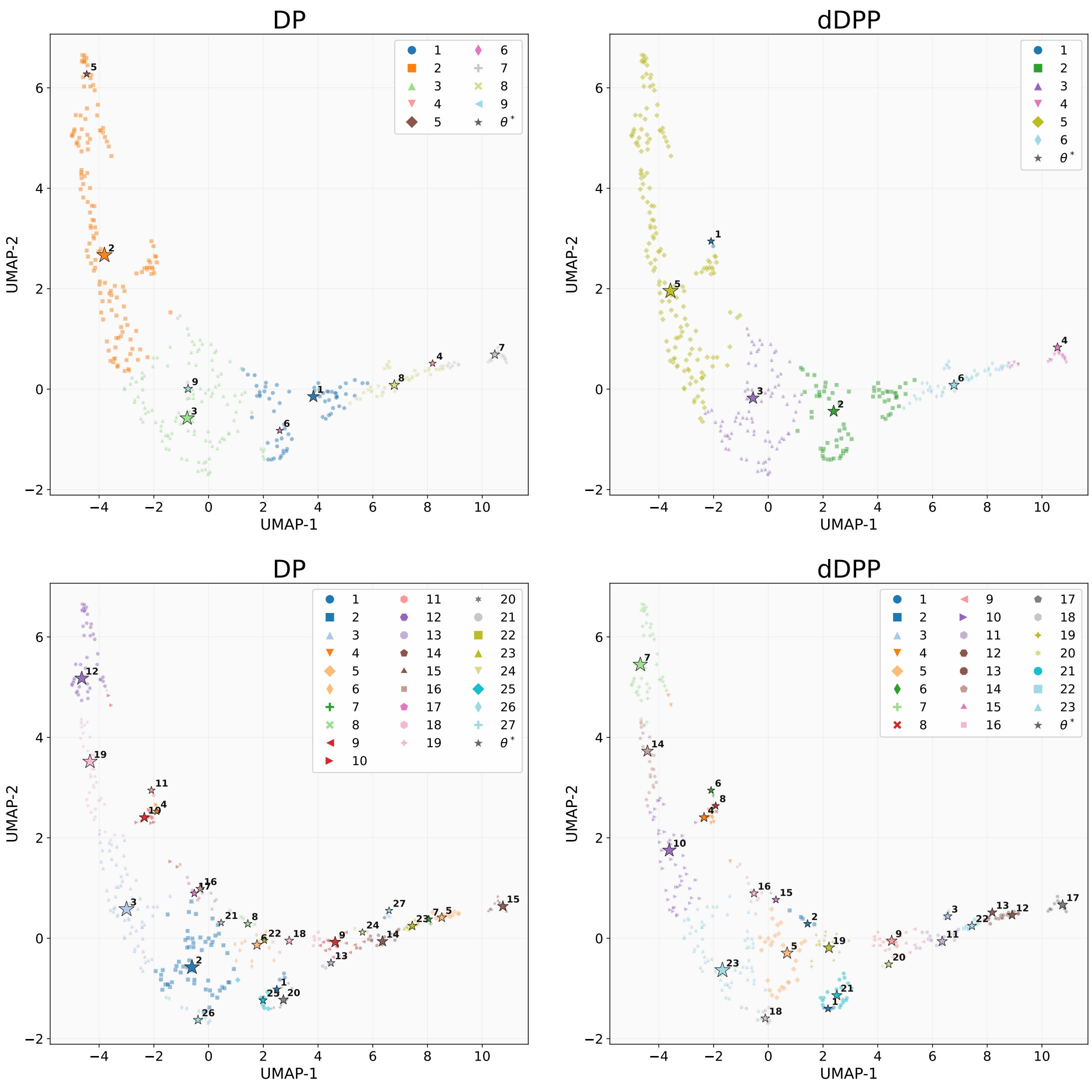



Figure 12: Same as Figure 3 with $w = 100$ (first row) and $w = 500$ (second row).

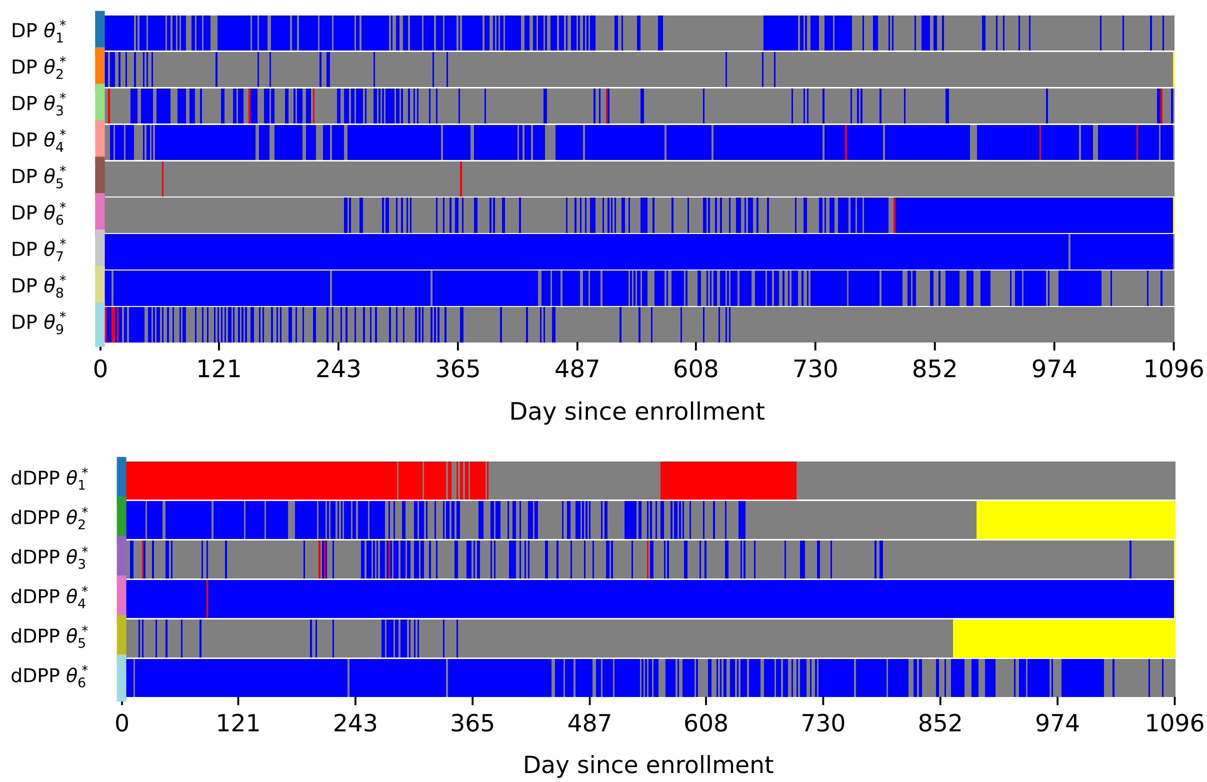



Figure 13: Same as Figure 4 with $w = 100$ for DP model (first row) and dDPP model (second row).

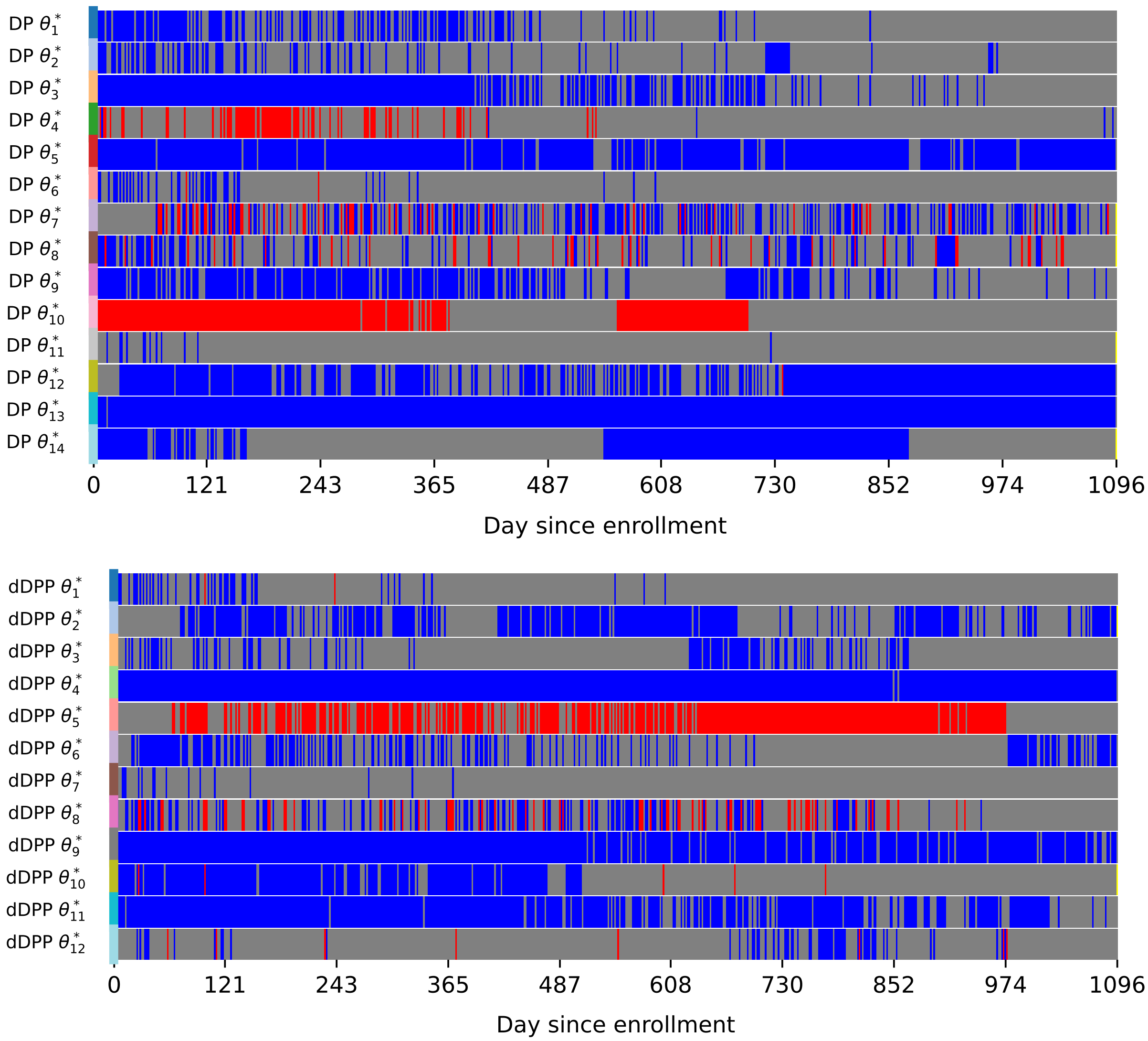
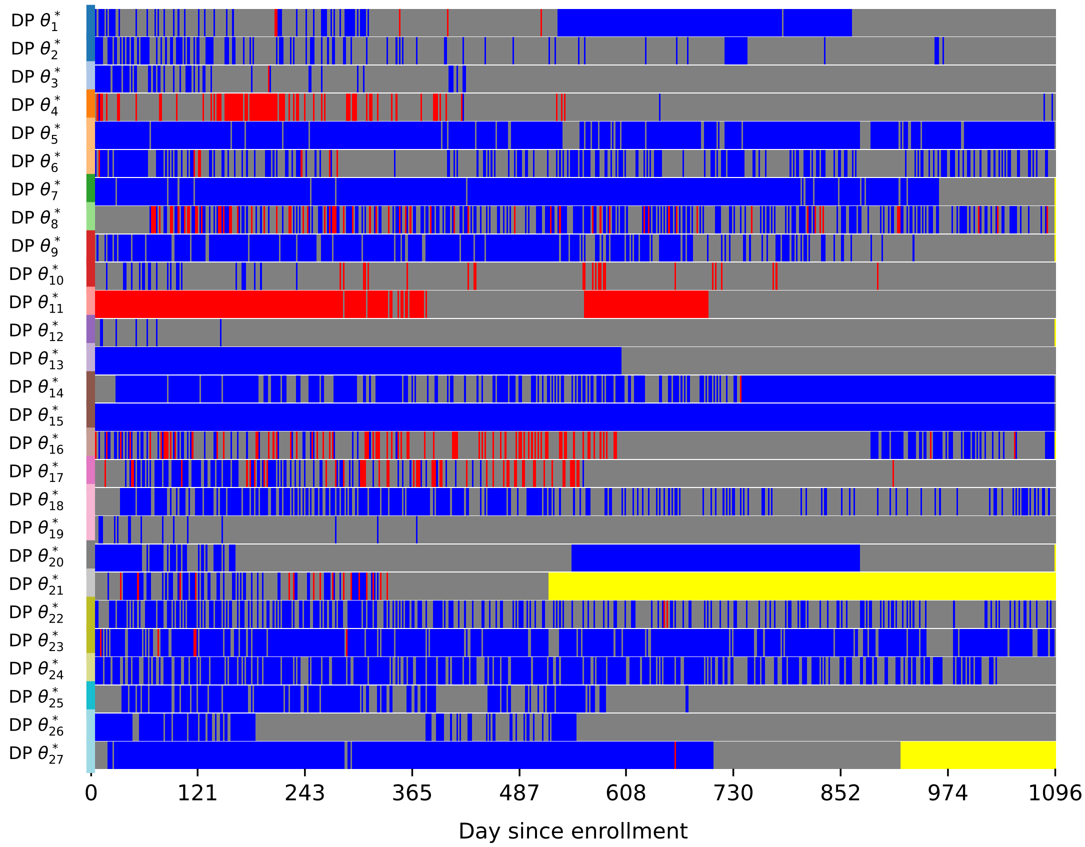



Figure 14: Same as Figure 4 with $w = 200, \gamma = 10$ for DP model (first row) and dDPP model (second row).



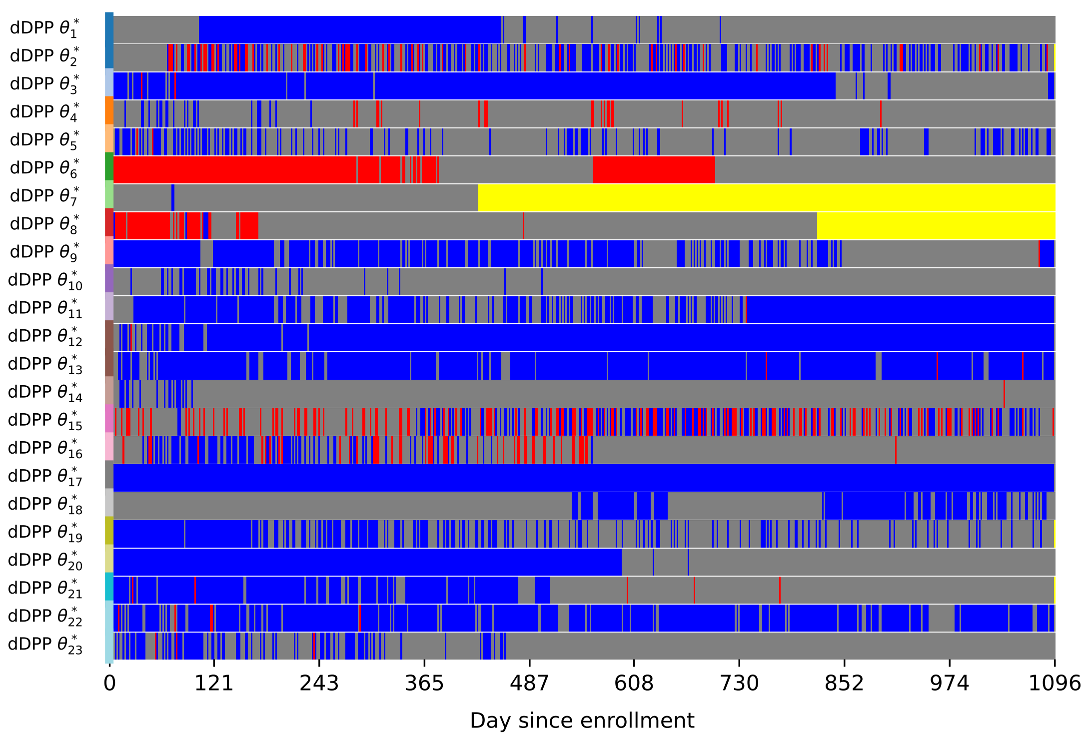




Figure 15: Same as Figure 4 with $w = 500$ for DP model (first row) and dDPP model (second row).